

Detection of low-energy tau neutrinos with the ANTARES neutrino telescope - a feasibility study

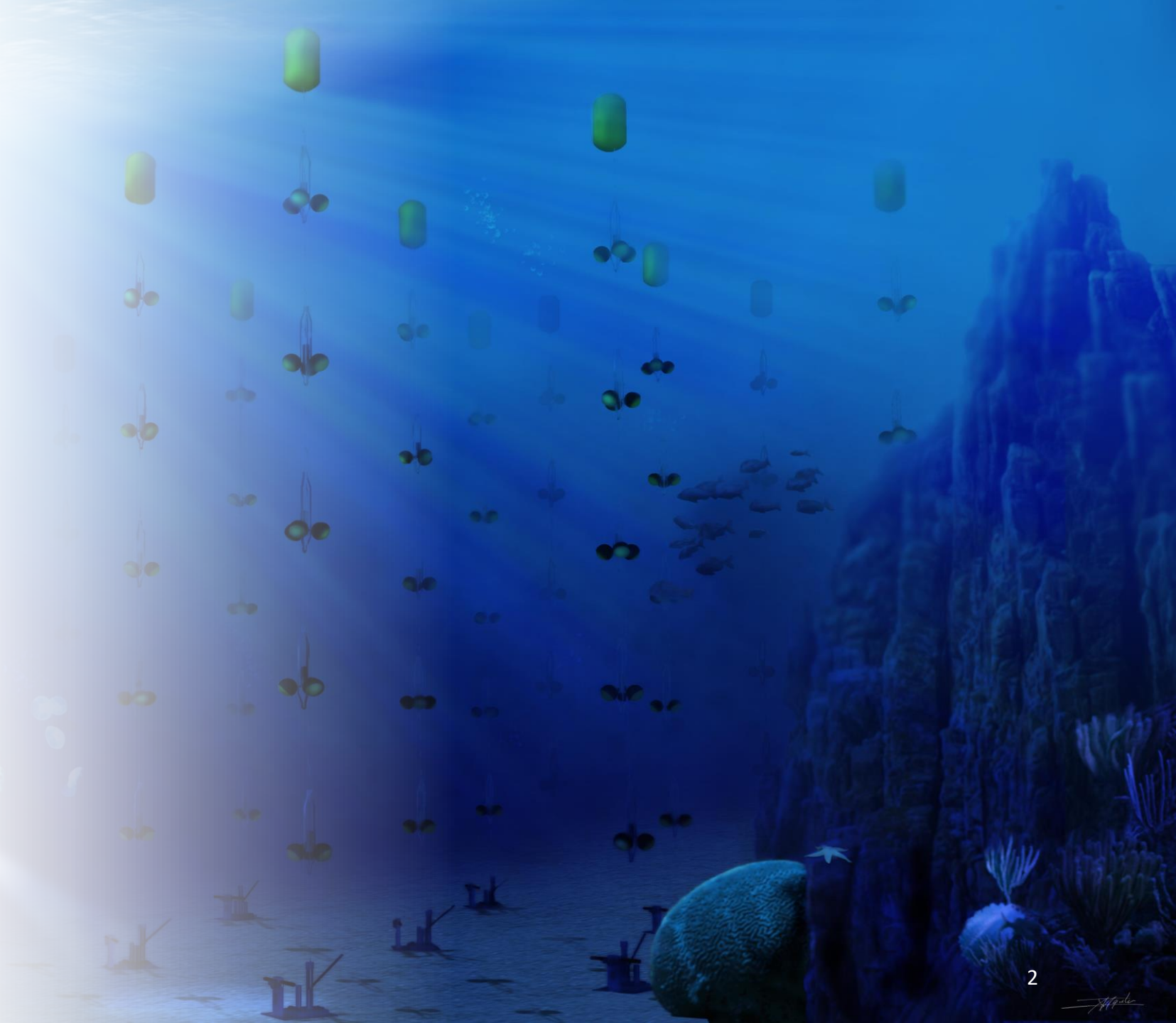
Michael Chadolias

Erlangen, 22nd of January 2025

M.Sc. Colloquium

Contents

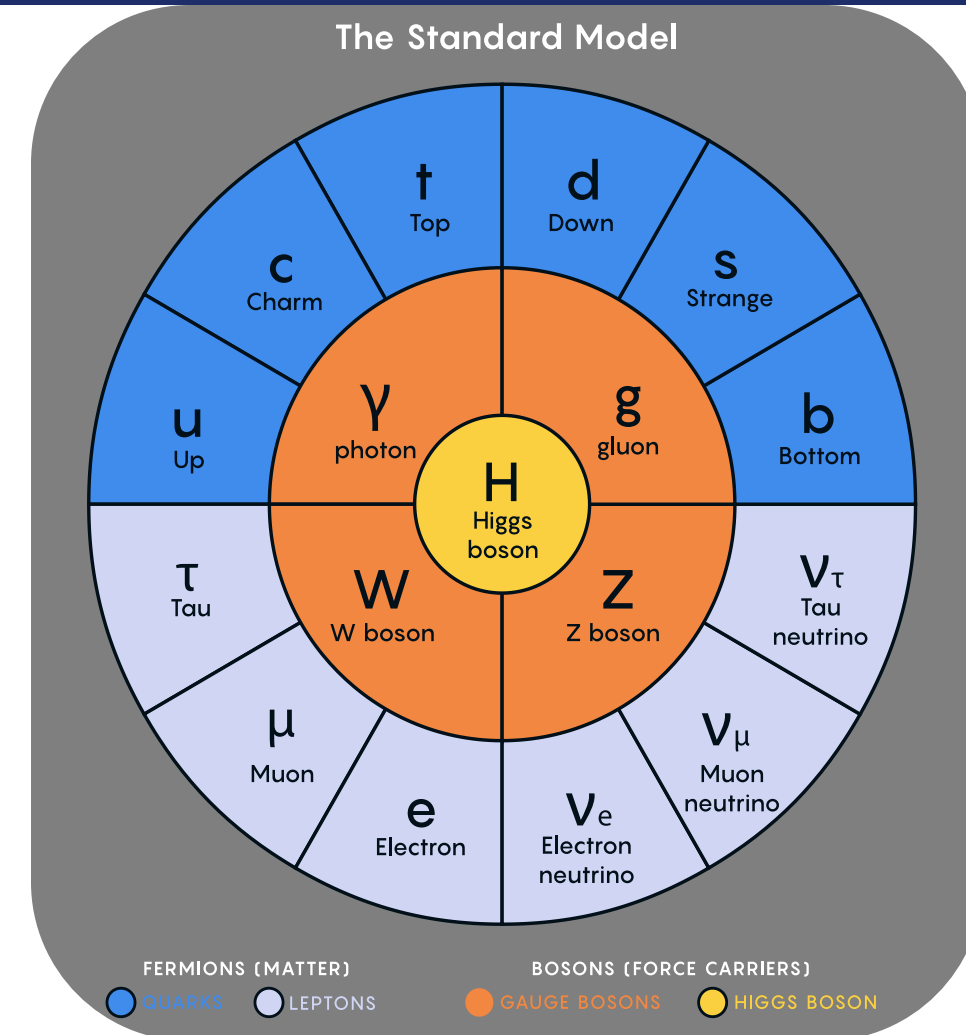
- 1) Neutrinos
- 2) ANTARES
- 3) Tau appearance measurement
- 4) ANTARES Monte-Carlo events
- 5) Detector's sensitivity to the tau appearance measurement
- 6) Summary & Outlook





Neutrinos

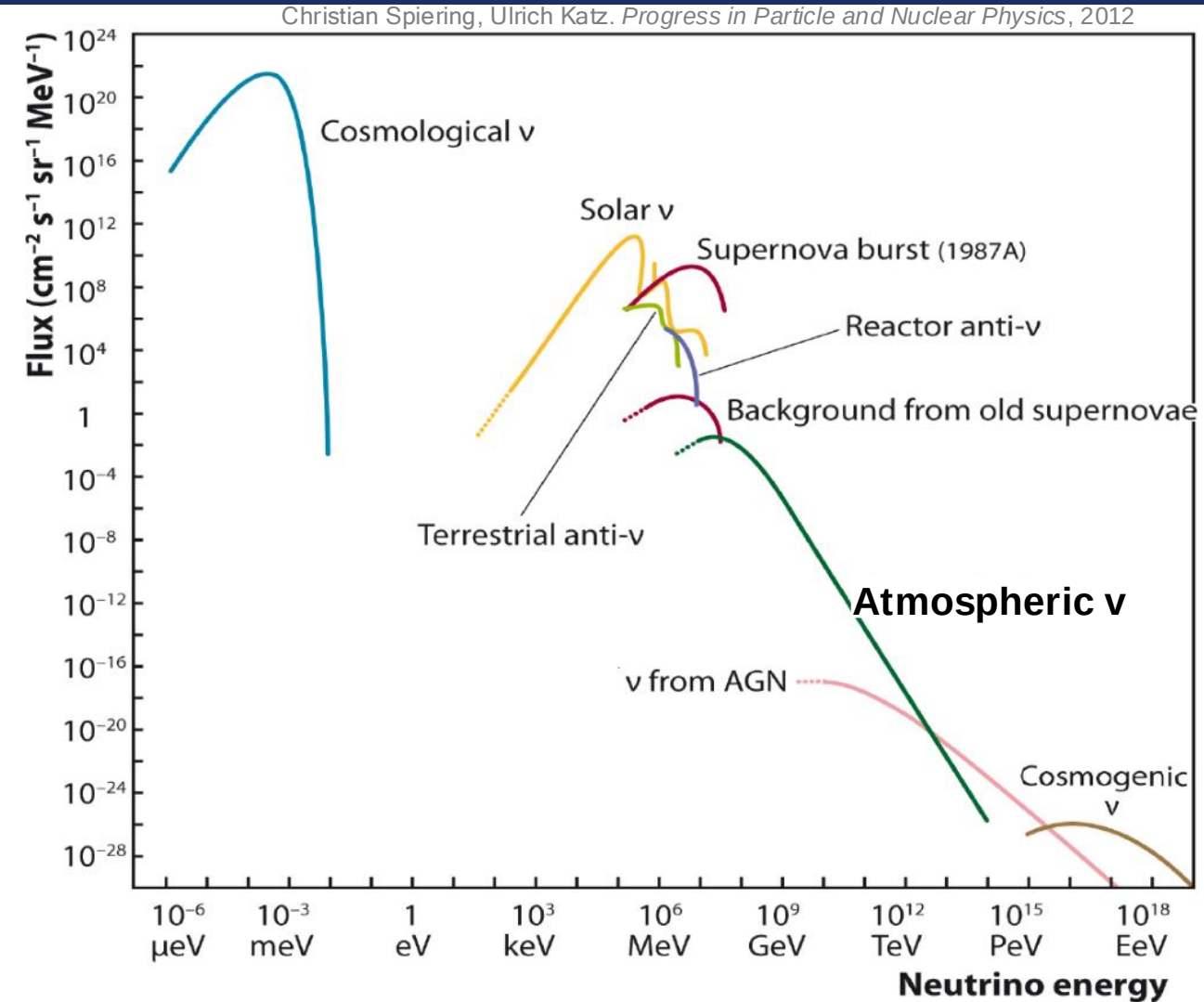
- Lightest particle of the fermionic family of the standard model of particle physics
- There are 3 known flavors
- Interacts under the following forces:
 - Weak nuclear force
 - Gravity
- Neutrinos can undergo oscillations transforming into another flavor



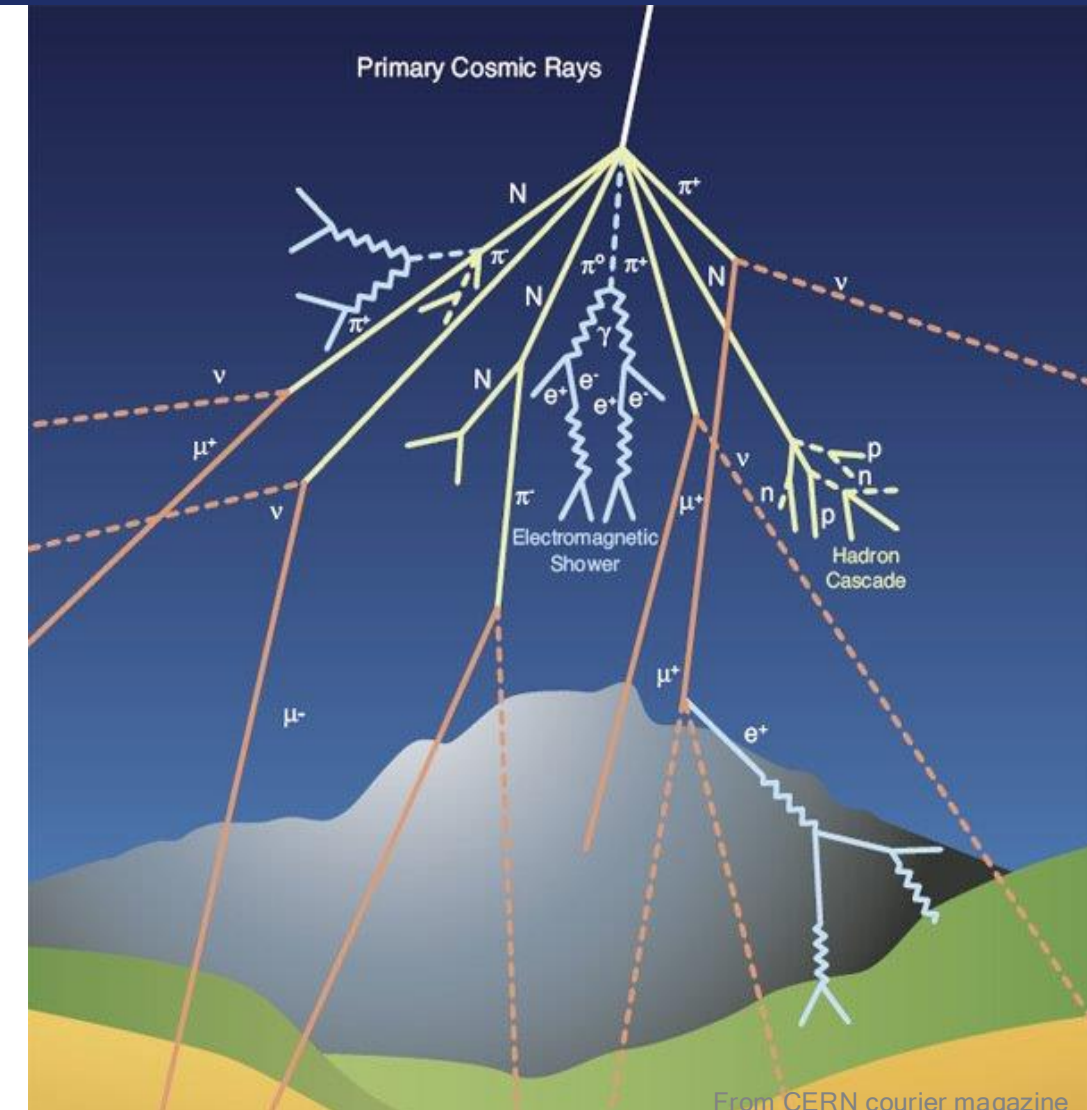
Neutrinos

Sources

- Sources are both natural and artificial
- Large energy range from μeV to EeV
- Ubiquity across scales from 10^{-28} to 10^{24}



- Originate from collisions of cosmic rays in Earth's atmosphere
 - Unstable kaons and pions are produced, decaying into:
$$\pi^+ / K^+ \rightarrow \mu^+ + \nu_\mu \quad \pi^- / K^- \rightarrow \mu^- + \bar{\nu}_\mu$$
$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu \quad \mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$
- Only **electron** and **muon** neutrinos are originally produced in the GeV energy range!
- How is a tau neutrino flux observed?
➡ **Neutrino Oscillations!**



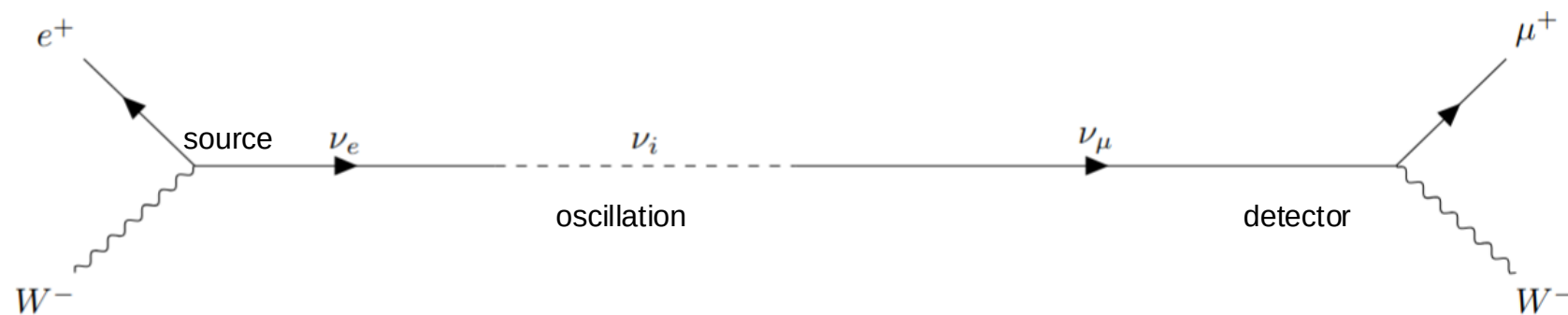
$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu1} & U_{\mu2} & U_{\mu3} \\ U_{\tau1} & U_{\tau2} & U_{\tau3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$

- Neutrino mixing matrix

$$|\nu_e\rangle = U_{e1}^* |\nu_1\rangle + U_{e2}^* |\nu_2\rangle + U_{e3}^* |\nu_3\rangle$$

- Phenomenology for the 3-flavor case defined with 4 parameters

- 3 mixing angles: θ_{ij}
- 1 CP-violating phase: δ_{ij}



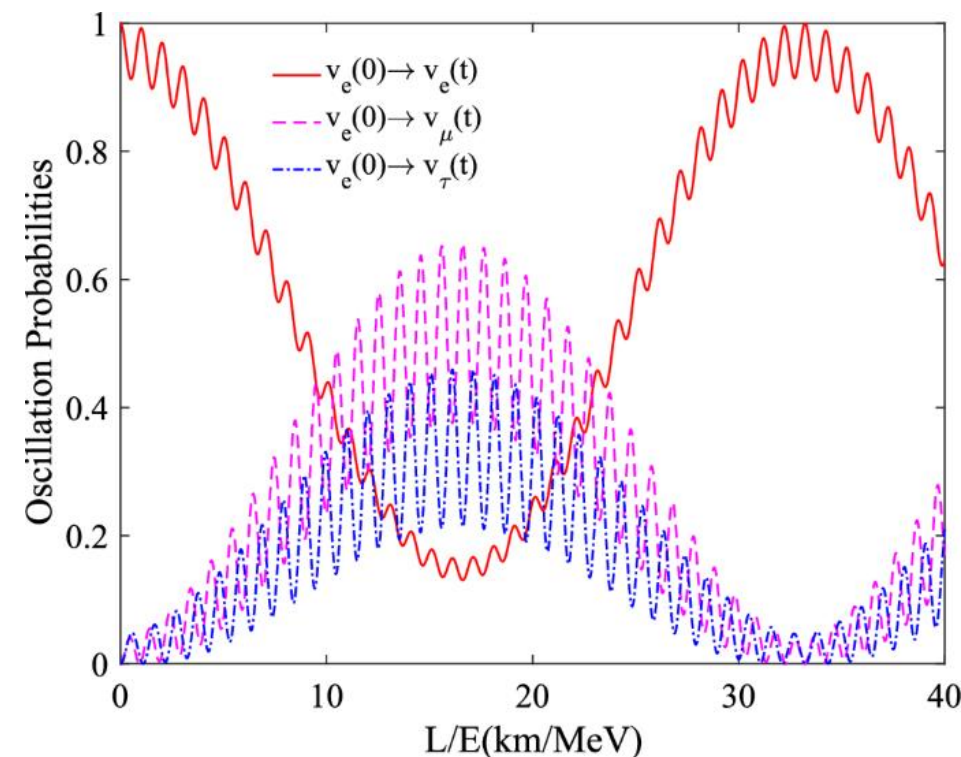
- Oscillation probability for a neutrino from flavor $\alpha \rightarrow \beta$ in vacuum

$$P(\nu_\alpha \rightarrow \nu_\beta) = \delta_{\alpha\beta} - 4 \sum Re[U_{\beta i} U_{\alpha i}^* U_{\beta j}^* U_{\alpha j}] \sin^2 \left(\frac{\Delta m_{ij}^2 L}{4E} \right) + 2 \sum Im[U_{\beta i} U_{\alpha i}^* U_{\beta j}^* U_{\alpha j}] \sin \left(\frac{\Delta m_{ij}^2 L}{4E} \right)$$

For a given L/E , oscillations are dominated by only one Δm^2 !

- Two flavor case:
 - Amplitude depends on: θ
 - Frequency depend on: $L/E, \Delta m^2$

$$P(\nu_\alpha \rightarrow \nu_\beta) = \underbrace{\sin^2(2\theta)}_{\text{Amplitude}} \underbrace{\sin^2 \left(\frac{\Delta m^2 L}{4E} \right)}_{\text{Frequency}}$$



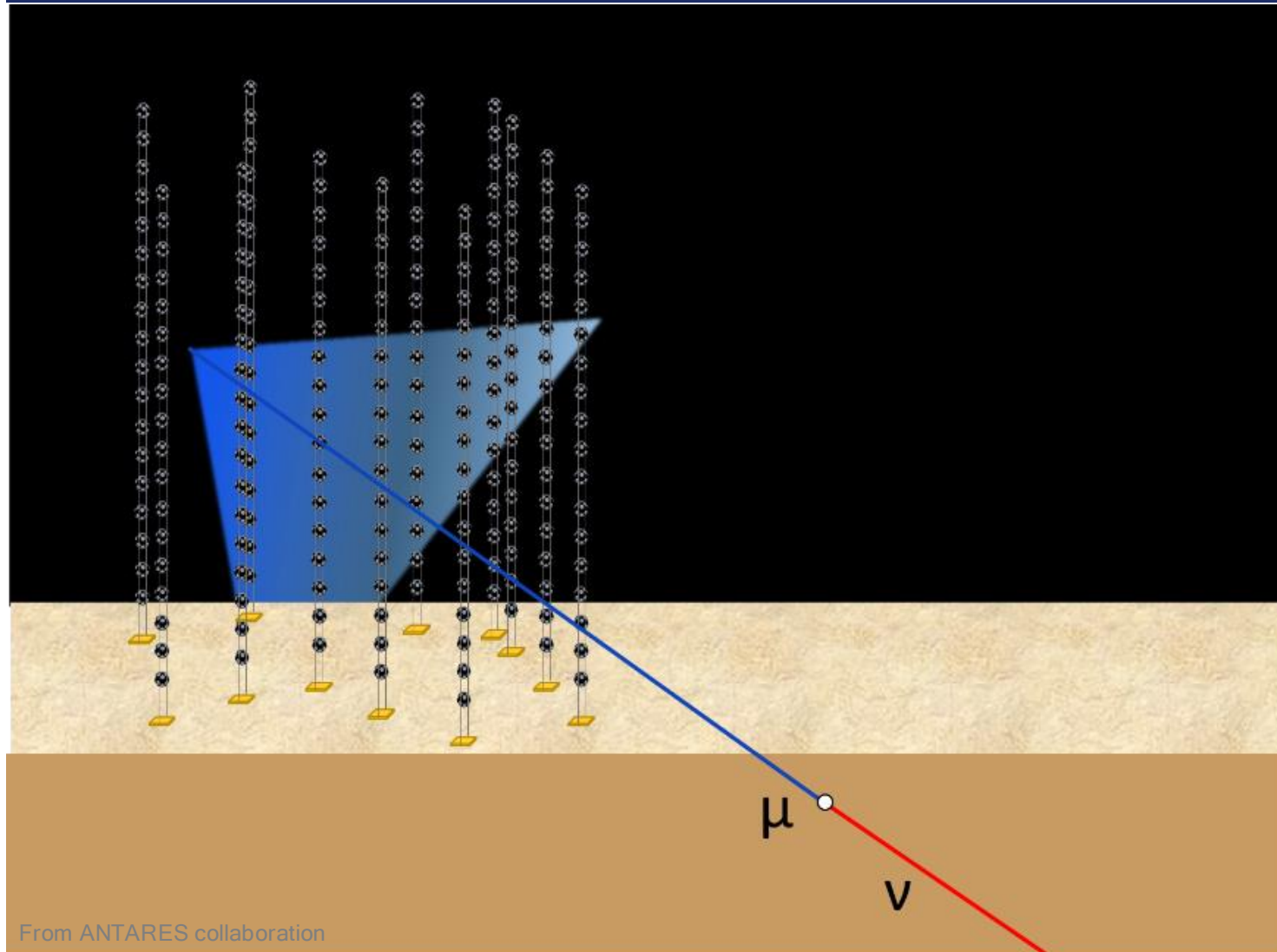
- Tau neutrinos are one of the least well measured particles in the standard model of particle physics
 - Reduce the uncertainties in the charged-current cross section
 - Investigate the unitarity of the neutrino mixing matrix
- } *Tau appearance measurement*
- Relevant results from OPERA, IceCube/DeepCore, Super-Kamiokande and KM3NeT
 - Is a potential combined analysis with KM3NeT possible?
 - ➡ First usage of ANTARES pseudo-data with KM3NeT tools!
 - Additional goal for ANTARES data beyond its main objective?
 - ➡ A preliminary feasibility analysis necessary



Neutrino Telescopes

Neutrino Telescopes

Introduction to Neutrino Telescopes



From ANTARES collaboration

Applications:

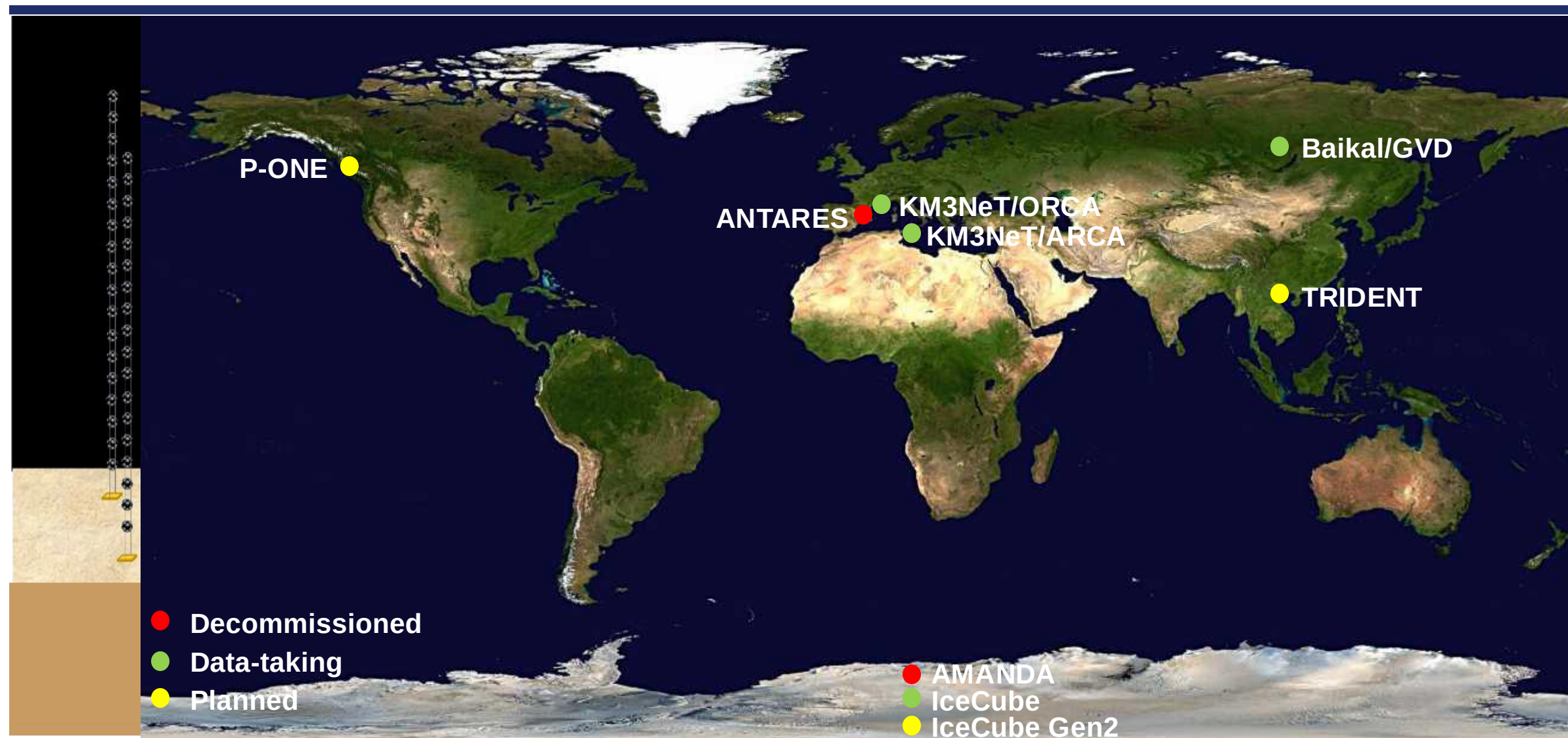
- Study of neutrino properties (e.g., oscillations).
- Investigation of astrophysical phenomena (e.g., supernovae, AGNs).
- Search for dark matter and exotic particles.

Neutrino Telescopes

Introduction to Neutrino Telescopes



ERLANGEN CENTRE
FOR ASTROPARTICLE
PHYSICS

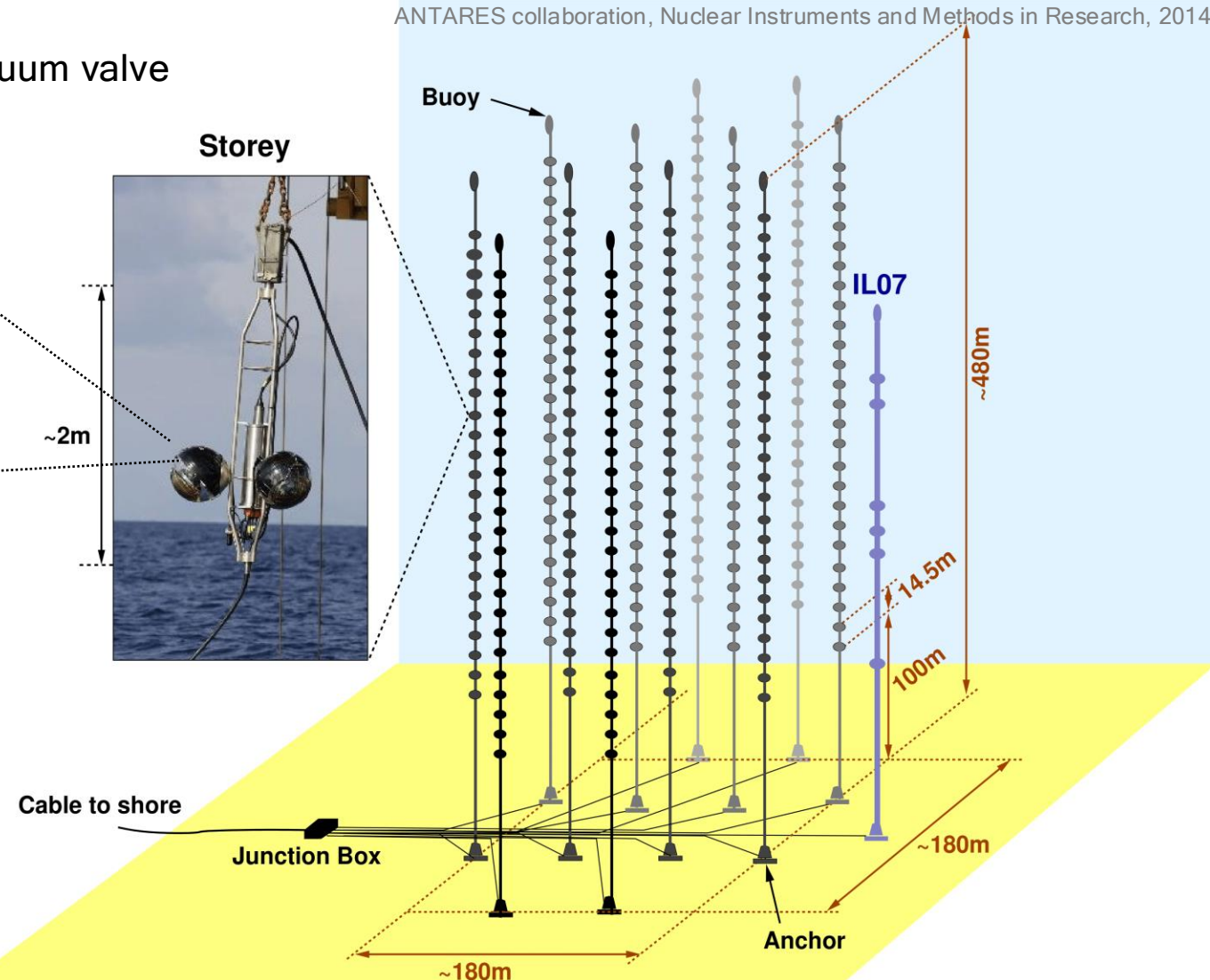


Neutrino Telescopes

ANTARES Detector

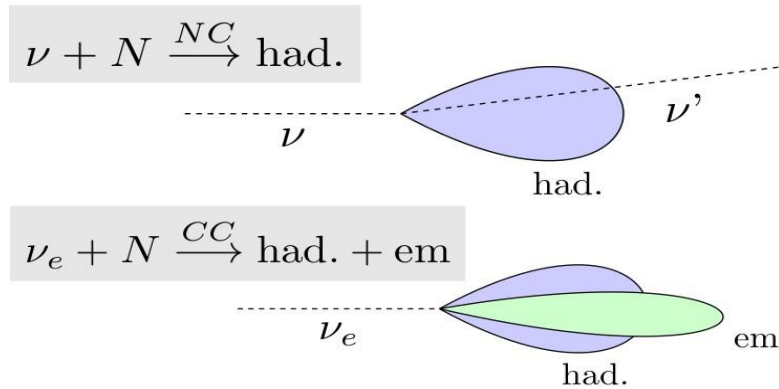


- Located 40 km off Toulon, France
- Sea depth: ~2500m
- Detector volume: ~0.03 km³
- Active: 2007 - 2022
- Energy threshold: 10 GeV

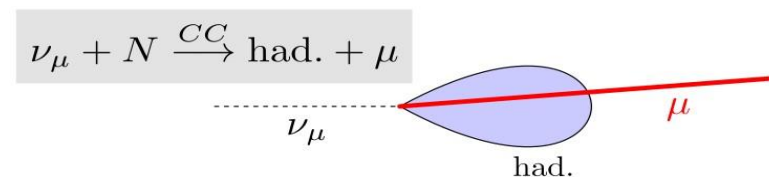


Non-tau channel

shower-like

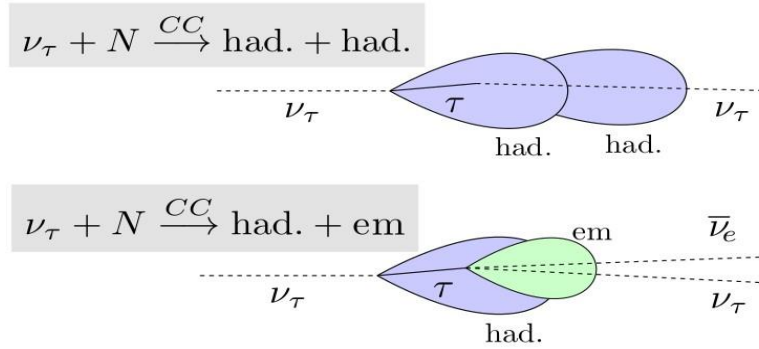


track-like

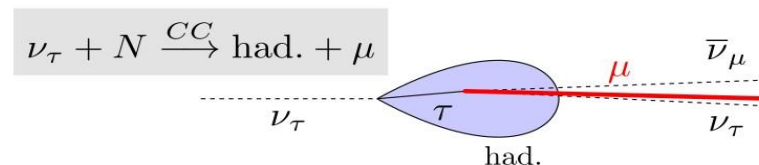


Tau channel

shower-like



track-like



- No event-by-event identification for tau neutrino interactions!
- Tau neutrinos result in **both tracks and showers** signatures!
- Event topology depends on the decay channel.

$\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau$	(17.8%)	} leptonic (35.2%)
$\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau$	(17.4%)	
$\tau^- \rightarrow \pi^- \pi^0 \nu_\tau$	(25.5%)	} hadronic (64.8%)
$\tau^- \rightarrow \pi^- \nu_\tau$	(10.8%)	
$\tau^- \rightarrow \pi^- \pi^0 \pi^0 \nu_\tau$	(9.3%)	
$\tau^- \rightarrow \pi^- \pi^- \pi^+ \nu_\tau$	(9.0%)	
$\tau^- \rightarrow \pi^- (> 2\pi) \nu_\tau$	(9.0%)	



Tau appearance

Tau events only product of neutrino oscillations:

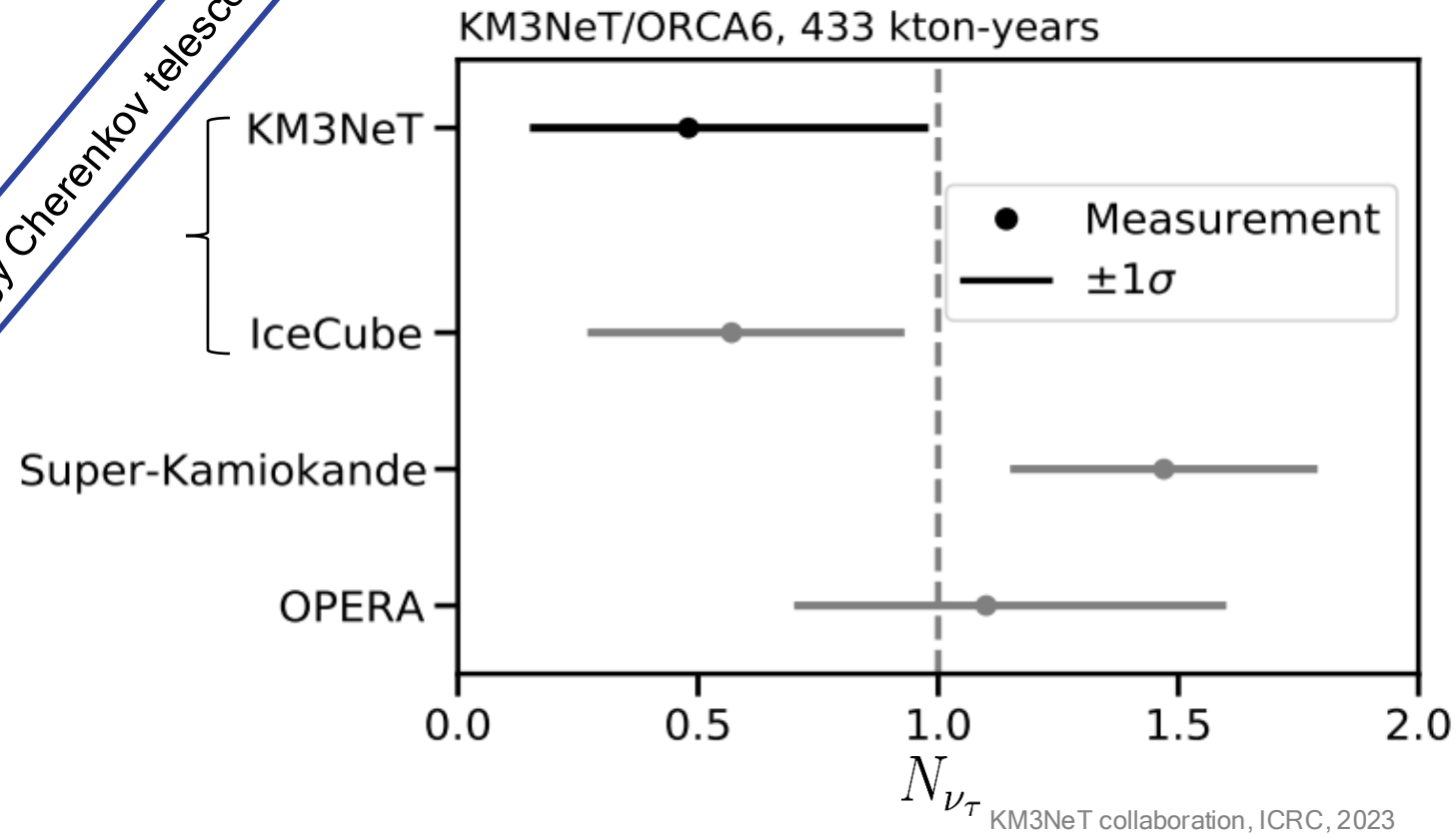
- deficit of muon neutrino events (track-like events)
- excess of shower events

$$N_{\nu_\tau} = \frac{N_{\tau, \text{measured}}}{N_{\tau, \text{expected}}}$$

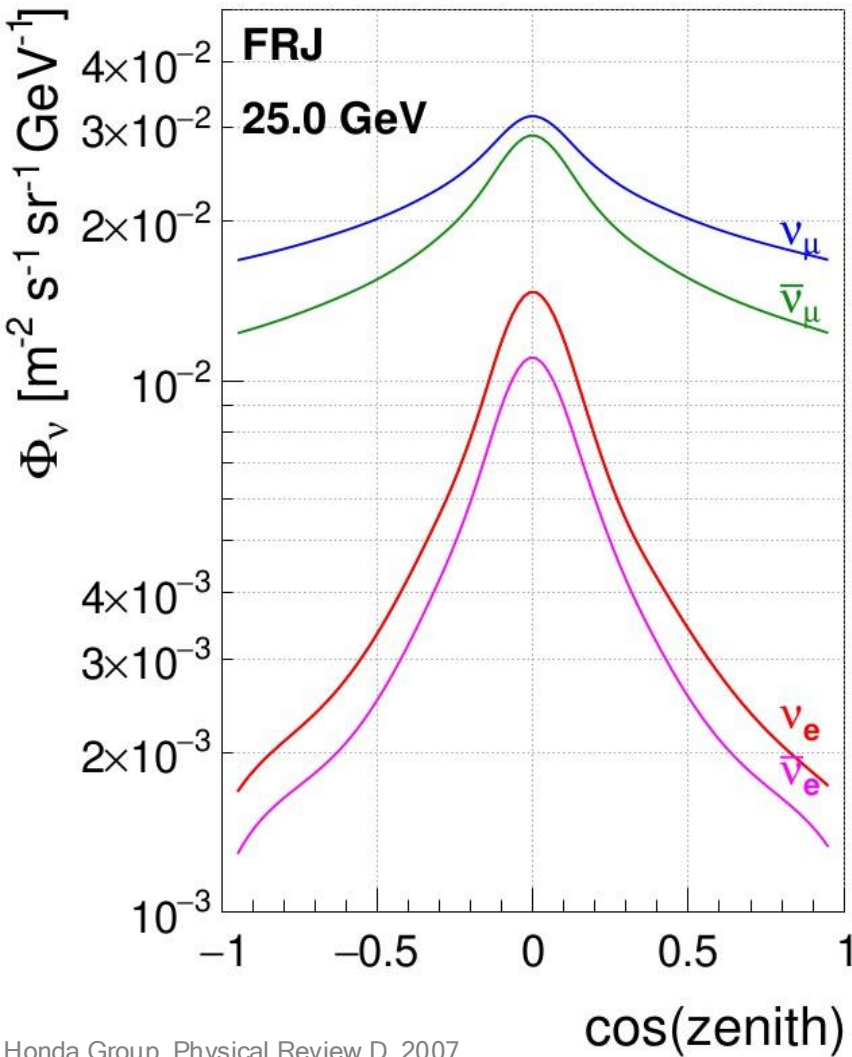
Tau normalization measurement:

- $N_{\nu_\tau} = 1$: expected case
- $N_{\nu_\tau} = 0$: no-tau case

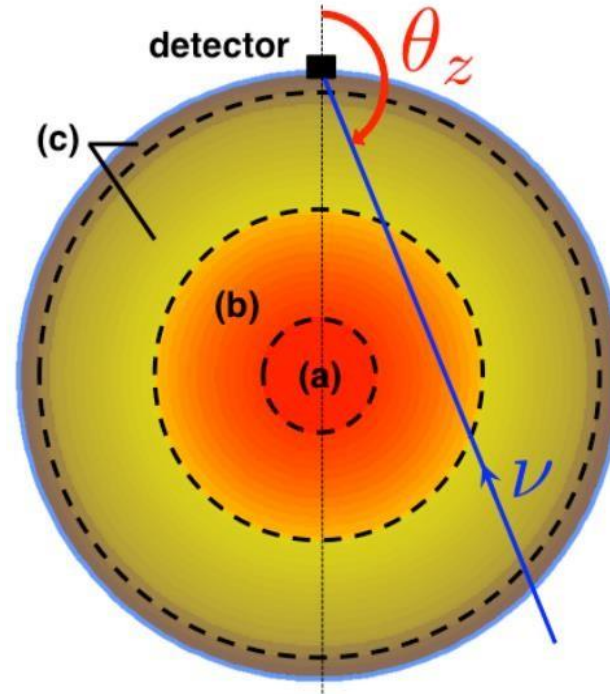
High-energy Cherenkov telescopes



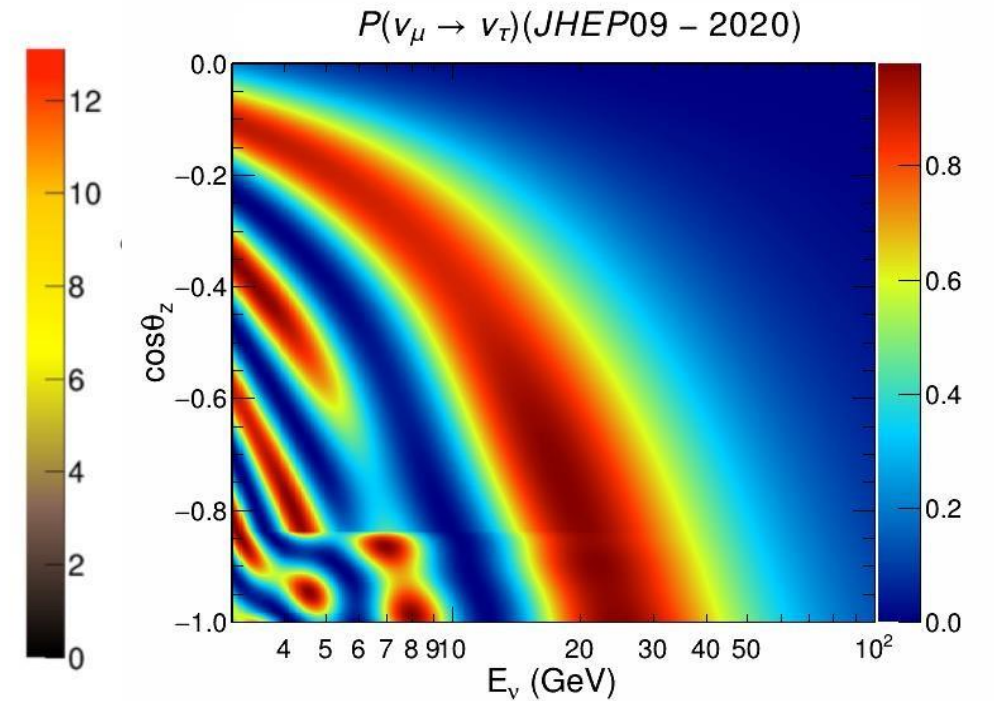
Tau Appearance



Honda Group, Physical Review D, 2007



Honda flux



Oscillation probability

Cross section

$$N_{\nu\beta}(E_\nu, \theta_\nu) = \Phi(E_\nu, \theta_\nu) \times P(\nu_\alpha \rightarrow \nu_\beta)(E_\nu, \theta_\nu) \times \sigma(E_\nu) \times \text{Detector Efficiency}$$

Tau Appearance

Sensitivity Calculation for the Tau Normalization Parameter



ERLANGEN CENTRE
FOR ASTROPARTICLE
PHYSICS



1 Fill 2D histograms (E, cos(θ)), for data and model

2 Obtain the negative log likelihood distribution, χ^2

$$\chi^2 = 2 \sum_{ij} \left(n_{ij}^{model} - n_{ij}^{data} + n_{ij}^{model} \log \left(\frac{n_{ij}^{data}}{n_{ij}^{model}} \right) \right) + \sum_{\alpha} \left(\frac{\alpha_{exp} - \alpha_{obs}}{\sigma_{\alpha}} \right)^2$$

Statistical part

Systematics part

Systematics

Oscillation parameters

Cross section

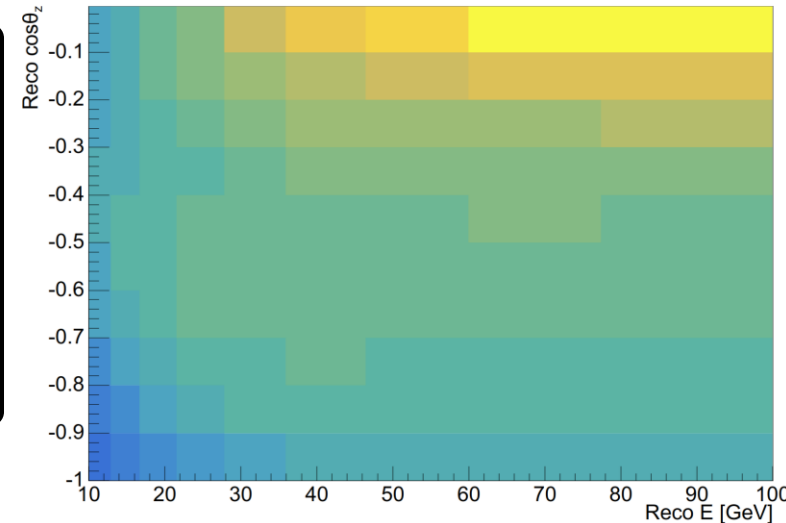
Background

Flux parameters

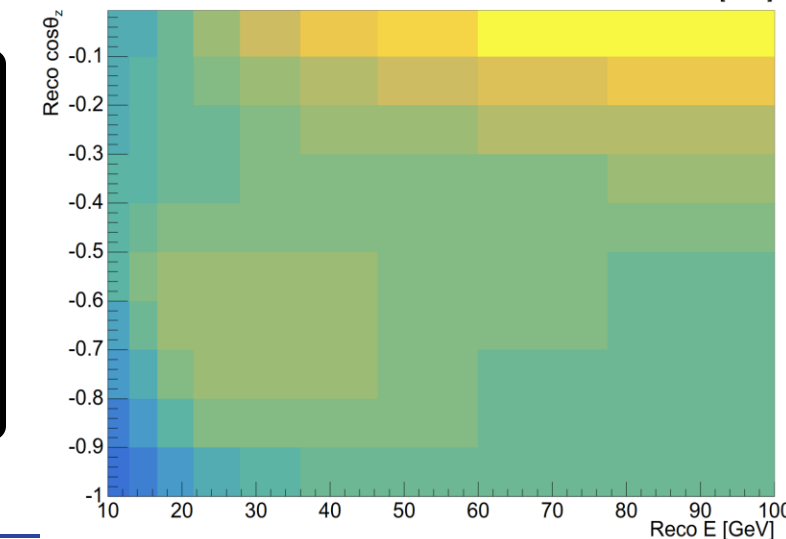
Detector response

A total of 17 parameters included!

Model Showers



Data Showers



Tau Appearance

Sensitivity Calculation for the Tau Normalization Parameter



ERLANGEN CENTRE
FOR ASTROPARTICLE
PHYSICS



3

Calculate the square difference between the hypothesis and best-fit model

$$\Delta\chi^2 = \chi_{\text{hypothesis}}^2 - \chi_{\text{best-fit}}^2$$

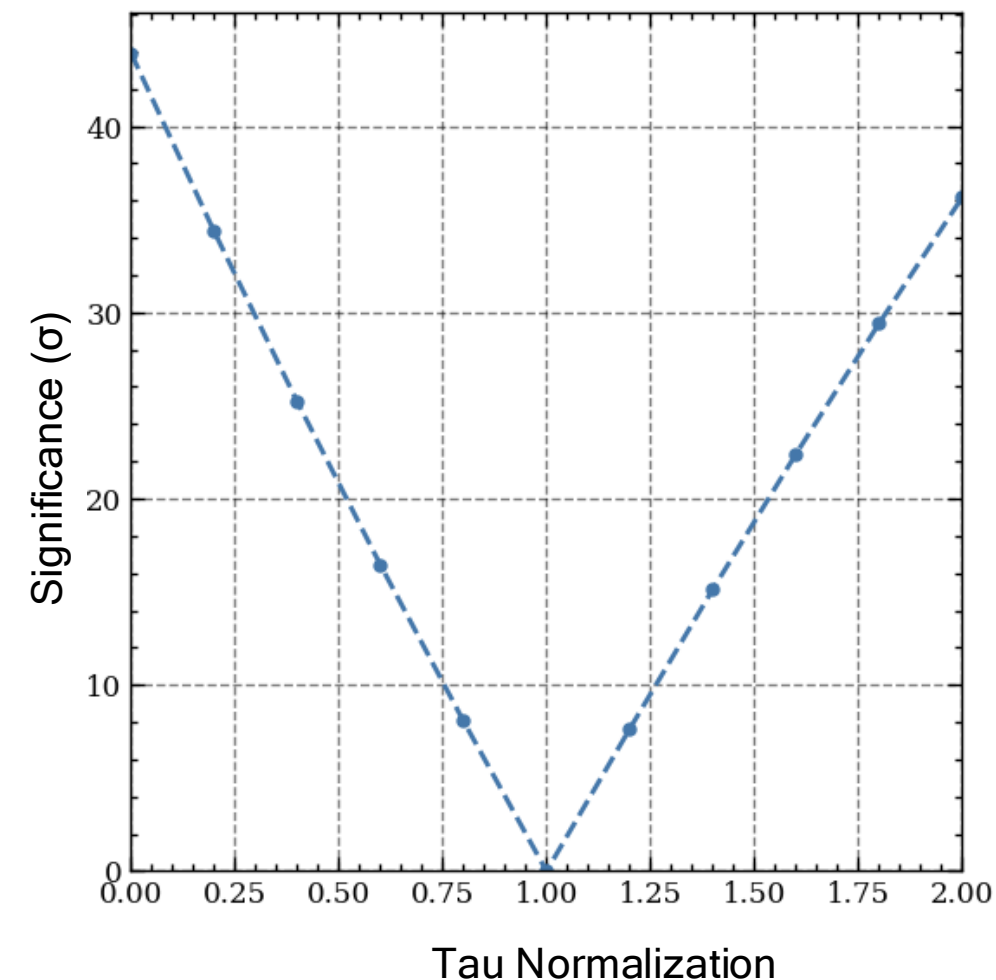
"best-fit":

Let the fit find the value of the Tau Normalization that corresponds to our data

"fixed hypothesis":

Calculate the likelihood of rejection for a certain fixed value of the Tau Normalization

$$\text{Significance approximated as } \sigma = \sqrt{\Delta\chi^2}$$





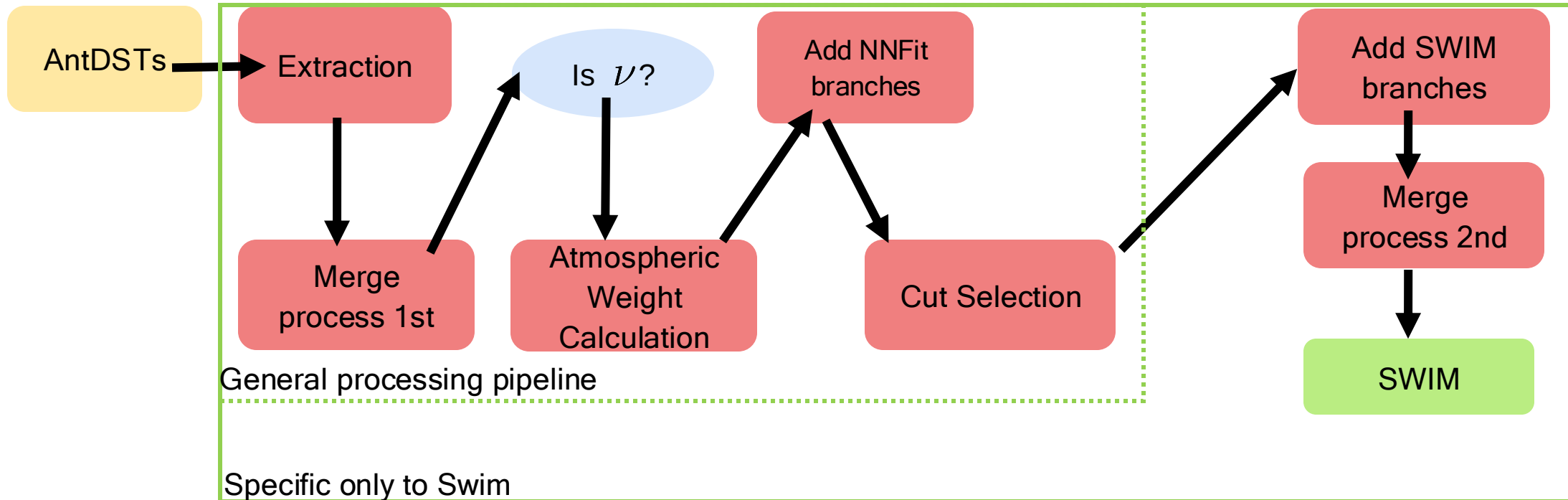
ANTARES Monte-Carlo

1 How many ν_τ are expected?

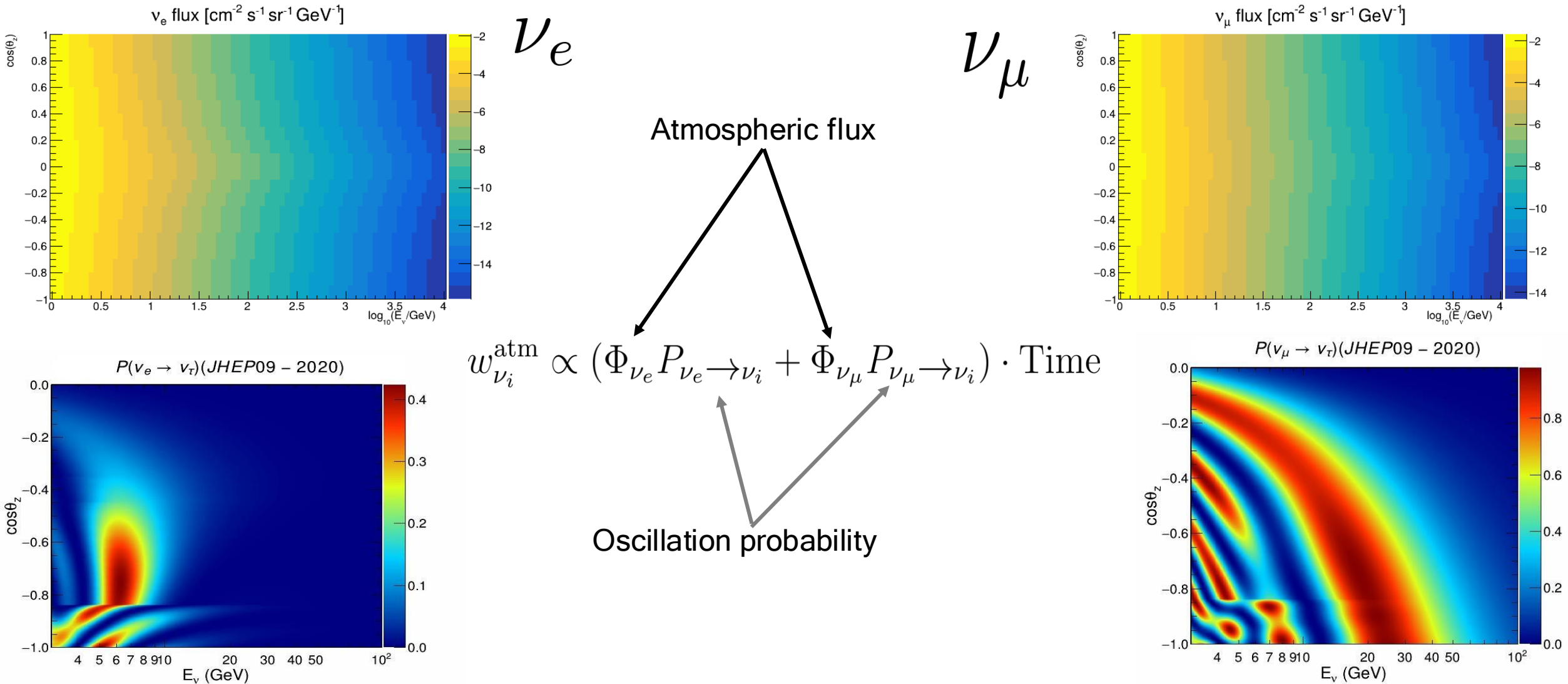
2 What is the performance of the reconstructions available in the AntDSTs, i.e. ANTARES summary file?

3 What is the performance of NNFit reconstruction?





- Applicable in both computing clusters (woody and cc lyon)
- Total computing time: around 4000 CPU hours
- First full lifetime ANTARES dataset with AntDST and NNFit information

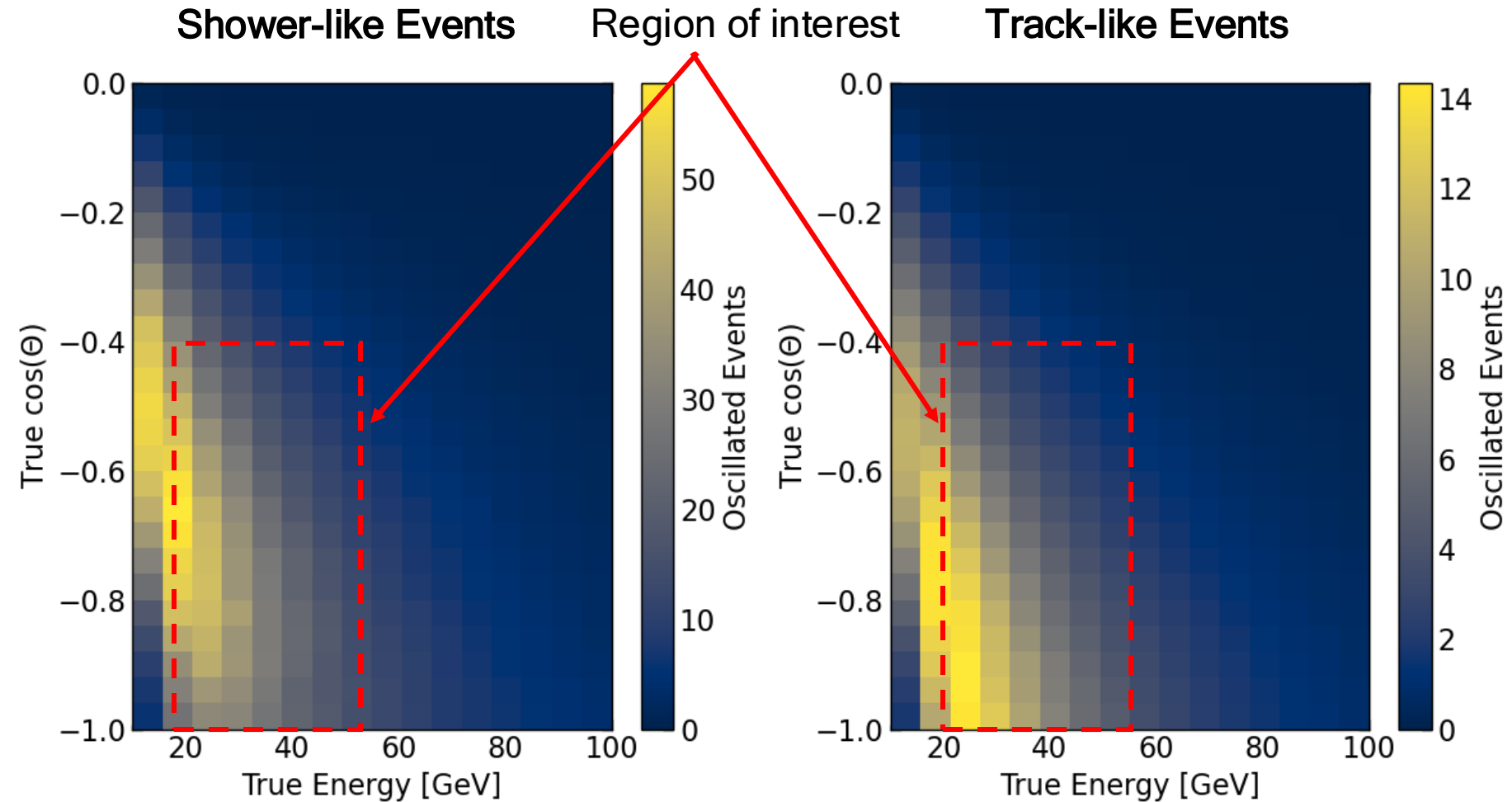


- Number of neutrinos for the full duration of the experiment
- Events are **triggered & reconstructed**
- Cuts:
 - $10 < E < 100 \text{ GeV}$
 - $\cos(\theta) < 0$
- Test case assuming perfect "muon" rejection

Category	Events	Percentage (%)
ν_e^{CC}	7773	13.5
ν_μ^{CC}	41023	71.5
ν_τ^{CCmu}	1059	1.8
$\nu_\tau^{\text{CCshower}}$	3355	5.8
ν^{NC}	4127	7.4
Total	57338	

Region of interest:

- Expected majority of events
- Maintains 90% of the sample
- Transition probability from $\nu_\mu \rightarrow \nu_\tau$ the highest

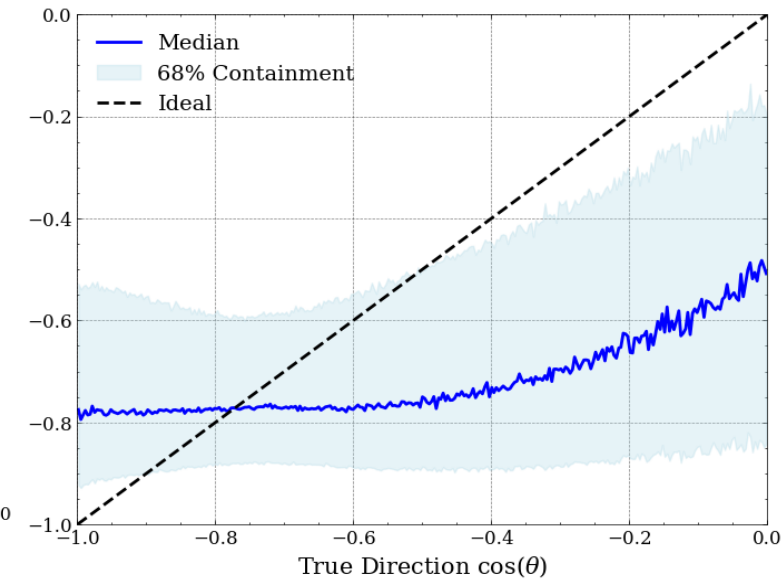
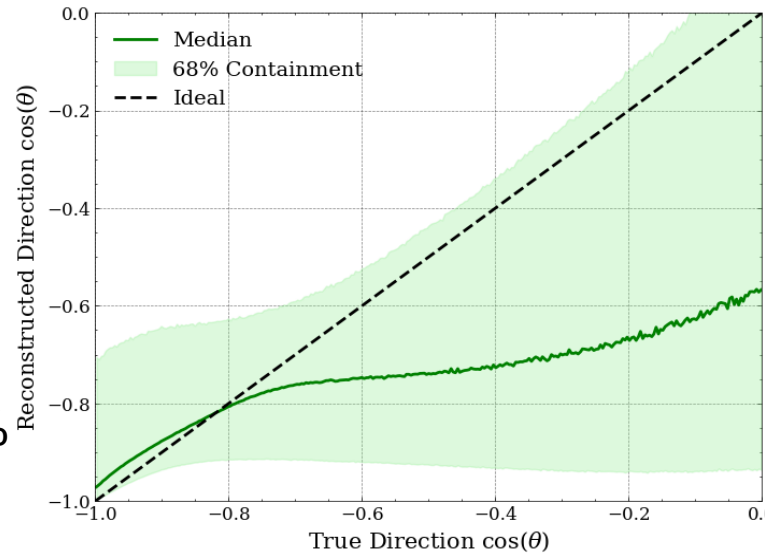


● Tracks

● Showers

AAFit:

- Maximum likelihood method
- Aims to reconstruct high energy events
- Advantages:
 - Reconstruction efficiency: almost 70%
 - Direction reconstruction for up-going tracks
- Disadvantages:
 - Direction reconstruction for showers

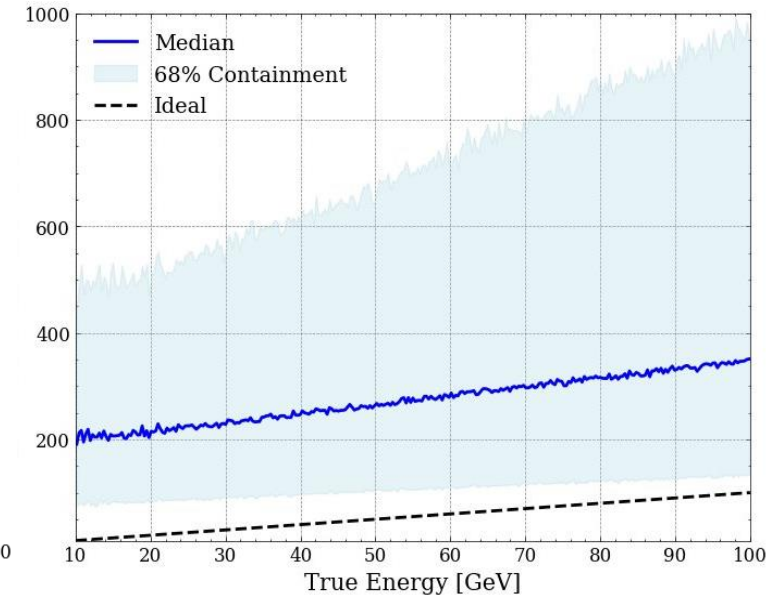
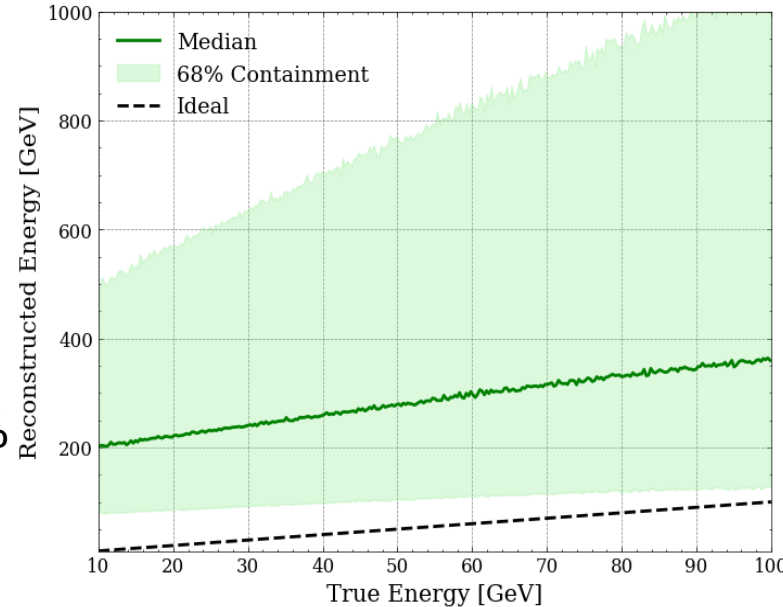


● Tracks

● Showers

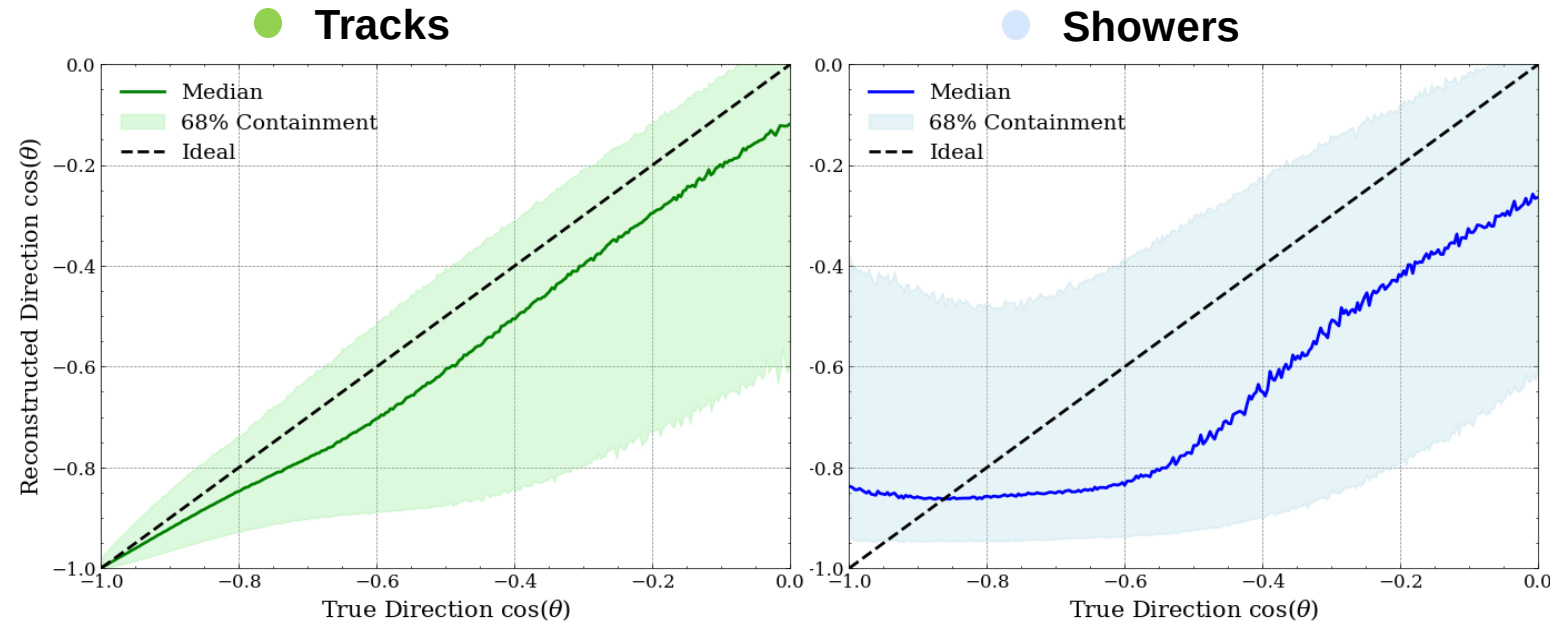
AAFit:

- Maximum likelihood method
- Aims to reconstruct high energy events
- Advantages:
 - Reconstruction efficiency: almost 70%
 - Direction reconstruction for up-going tracks
- Disadvantages:
 - Direction reconstruction for showers
 - Energy reconstruction far from ideal!
 - Efficiency drop to 10% even with loose quality cuts



BBFit:

- Maximum likelihood method
- Aims to reconstruct low energy track-like events
- Approximation for shower also exists
- **Advantages:**
 - Direction reconstruction for both topologies better than AAFit
 - Efficiency after quality cuts up to 15%
- **Disadvantages:**
 - Low reconstruction efficiency up to 30%



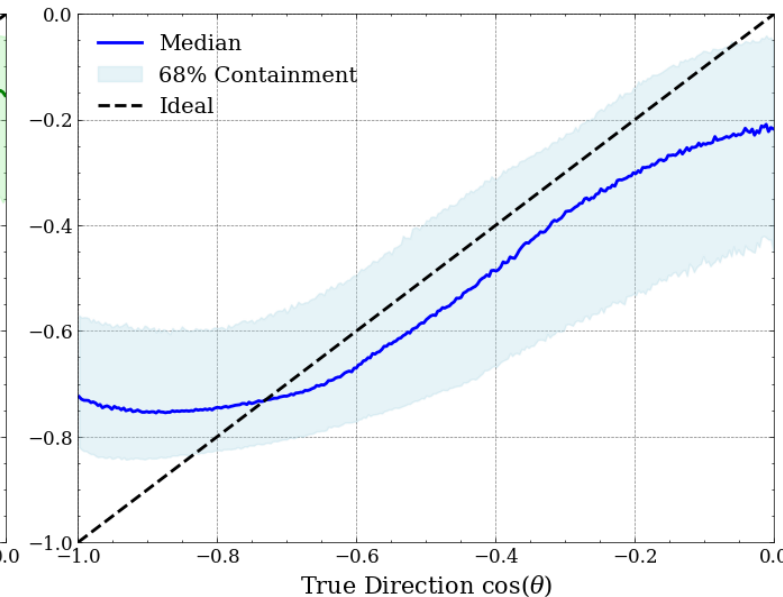
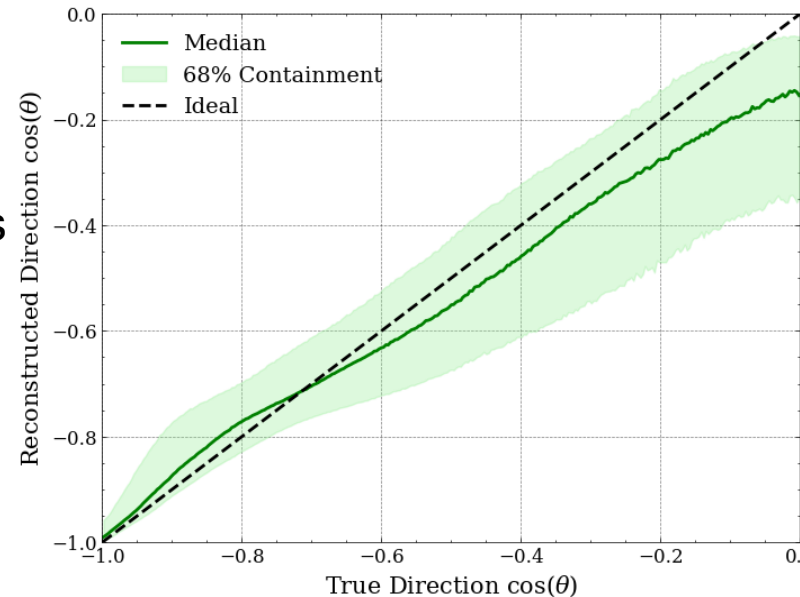
Direction performance: simple quality cuts

● Tracks

● Showers

NNFit:

- ML based method
- Aims to reconstruct low energy events
- Two networks developed for both topologies: NNFitTrack, NNFitShower
- **Advantages:**
 - Direction reconstruction for both topologies better than AAFit
 - Efficiency after quality cuts up to 15%
- **Disadvantages:**
 - Low reconstruction efficiency up to 25%



Direction performance: simple quality cuts

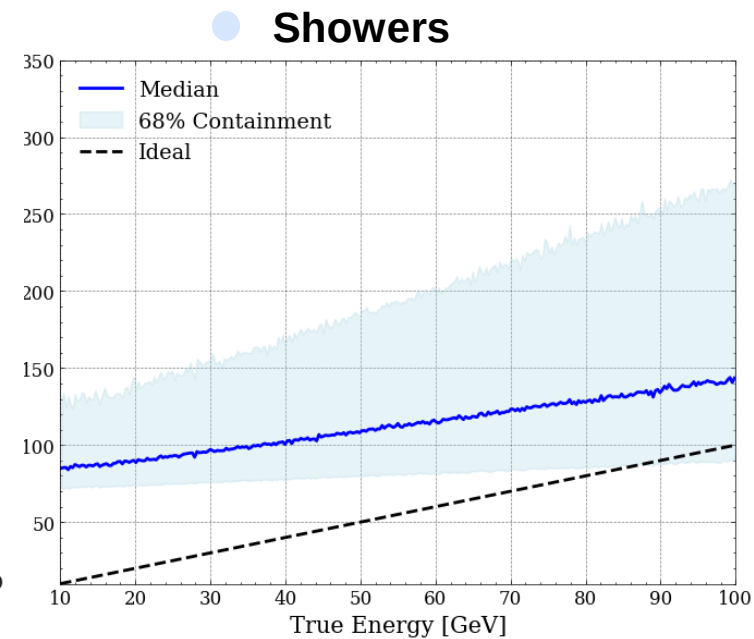
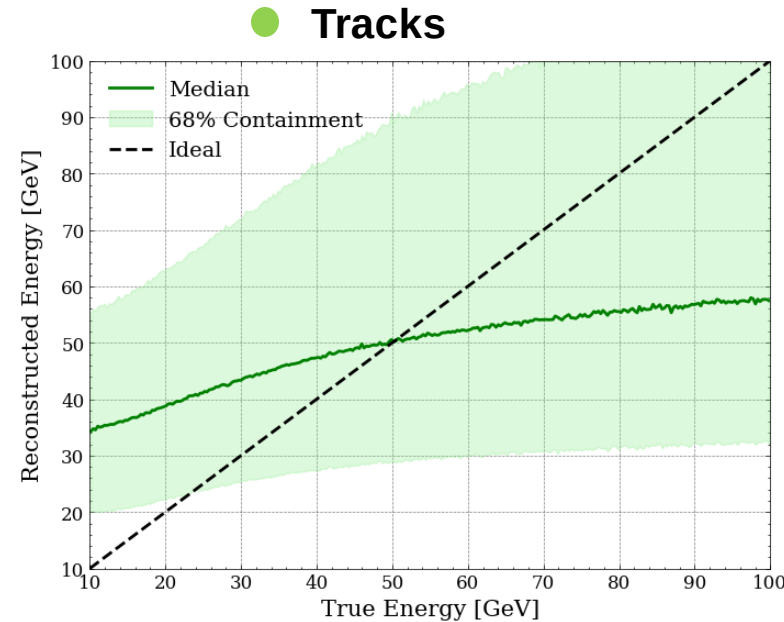


More information about the reconstruction

J.G. Mendez, S. Ardid, and M. Ardid., ICRC, 2023

NNFit:

- ML based method
- Aims to reconstruct low energy events
- Two networks developed for both topologies: NNFitTrack, NNFitShower
- **Advantages:**
 - Direction reconstruction for both topologies better than AAFit
 - Efficiency after quality cuts up to 15%
- **Disadvantages:**
 - Low reconstruction efficiency up to 25%
 - Energy reconstruction also not ideal



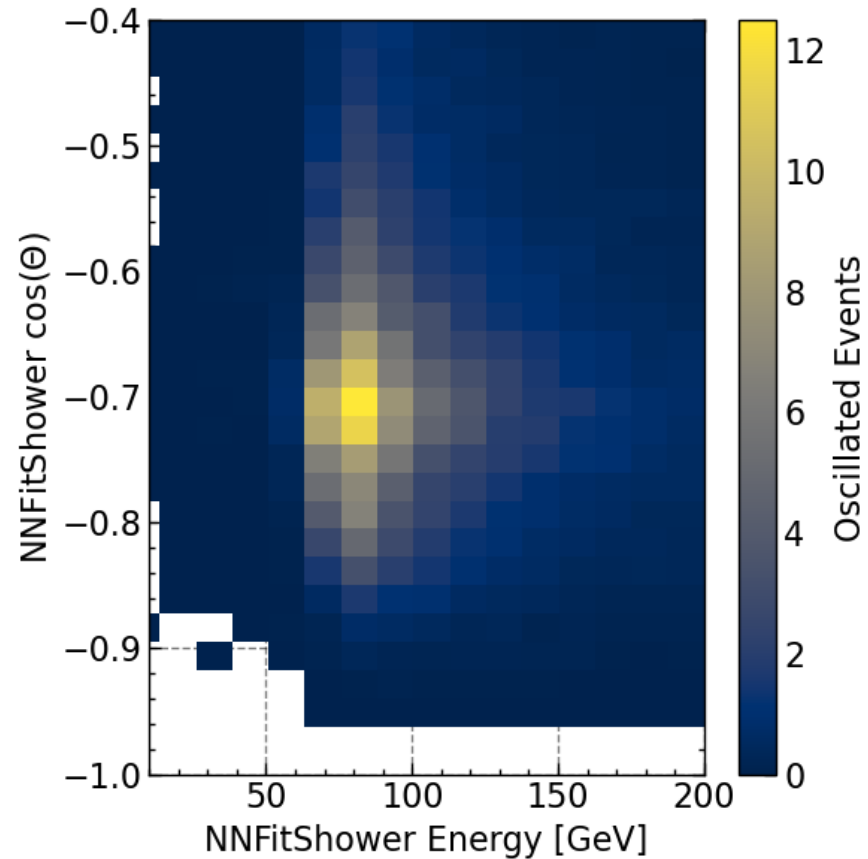
Energy performance: simple quality cuts

- Number of neutrinos for the full duration of the experiment reconstructed by NNFit with some quality cuts applied
- First iteration of selection criteria, not optimized!
- Assume "perfect" classification
- Cut selection for both topologies
- Tracks:
 - $\sigma_\theta < 8^\circ$
 - $\sigma_R < 10 \text{ m}$
 - $\sigma_Z < 10 \text{ m}$
 - $\cos(\theta) < -0.4$
- Showers:
 - $\sigma_R < 10 \text{ m}$
 - $\sigma_Z < 10 \text{ m}$
 - $\cos(\theta) < -0.4$

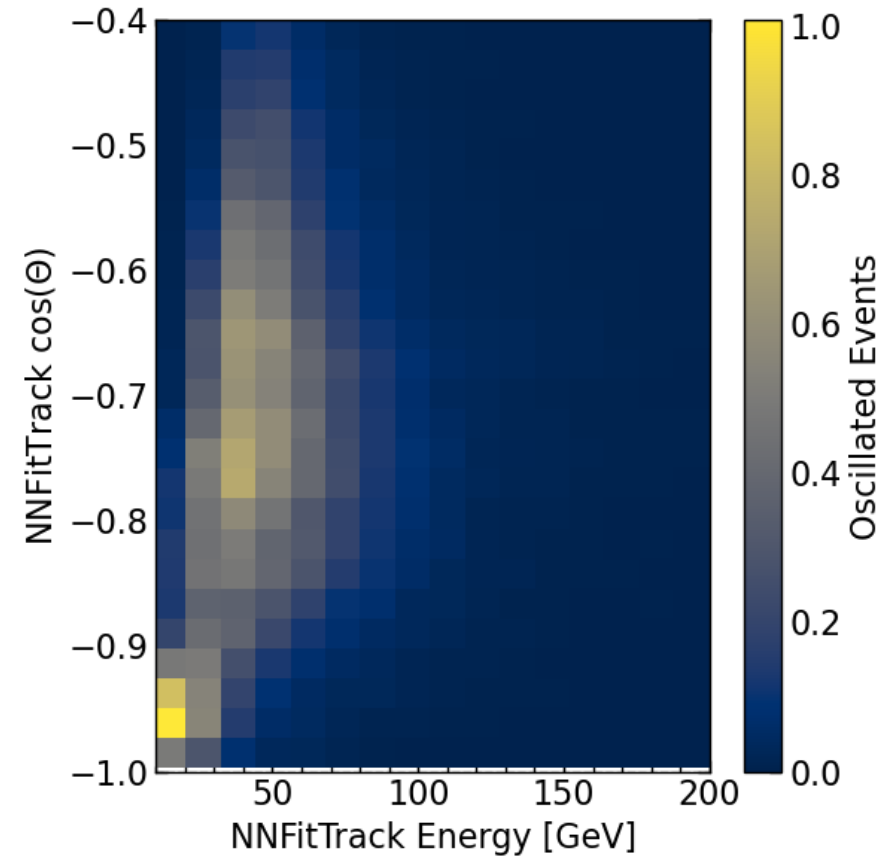
Category	Events	Percentage (%)
ν_e^{CC}	885	13.0
ν_μ^{CC}	3107	45.2
ν_τ^{CCmu}	67	0.9
$\nu_\tau^{\text{CCshower}}$	372	5.6
ν^{NC}	447	6.5
μ	1982	28.8
Total	6860	

Cut selection focused mainly on the predicted uncertainties from the network!

Shower-like Events



Track-like Events



	AAFit	BBFit	NNFit
Initial efficiency			
Efficiency after cuts			
Energy - Showers		No Data	
Energy - Tracks		No Data	
Direction - Showers			
Direction - Tracks			



Tau Appearance Sensitivity

Sensitivity calculations/scenarios depends on the following, parameters:

- Cut Selection
- Mass Ordering
- Interaction Channel
- Reconstruction
- Systematic Treatment

In total 120 different scenarios!

All sensitivities computed via the *Swim* software!

Detector response

Cut Selection

- True energy
- NNFit reconstructed upgoing
- NNFit reconstructed with quality cuts

Reconstruction

- MC truth
- AAFit
- NNFit

Model parameters

Mass Ordering

- Normal (NO)
- Inverted (IO)

Interaction Channel

- CC
- CC + NC

Systematic Treatment

- fixed
- fitted

Sensitivity calculations/scenarios depends on the following, parameters:

- Cut Selection
- Mass Ordering
- Interaction Channel
- Reconstruction
- Systematic Treatment

In total 120 different scenarios!

All sensitivities computed via the *Swim* software!

Scenarios presented here!

Cut Selection

- True energy
- NNFit reconstructed upgoing
- NNFit reconstructed with quality cuts

Reconstruction

- MC truth
- AAFit
- NNFit

Mass Ordering

- Normal (NO)
- Inverted (IO)

Interaction Channel

- CC
- CC + NC

Systematic Treatment

- fixed
- fitted

Tau Appearance - Sensitivity

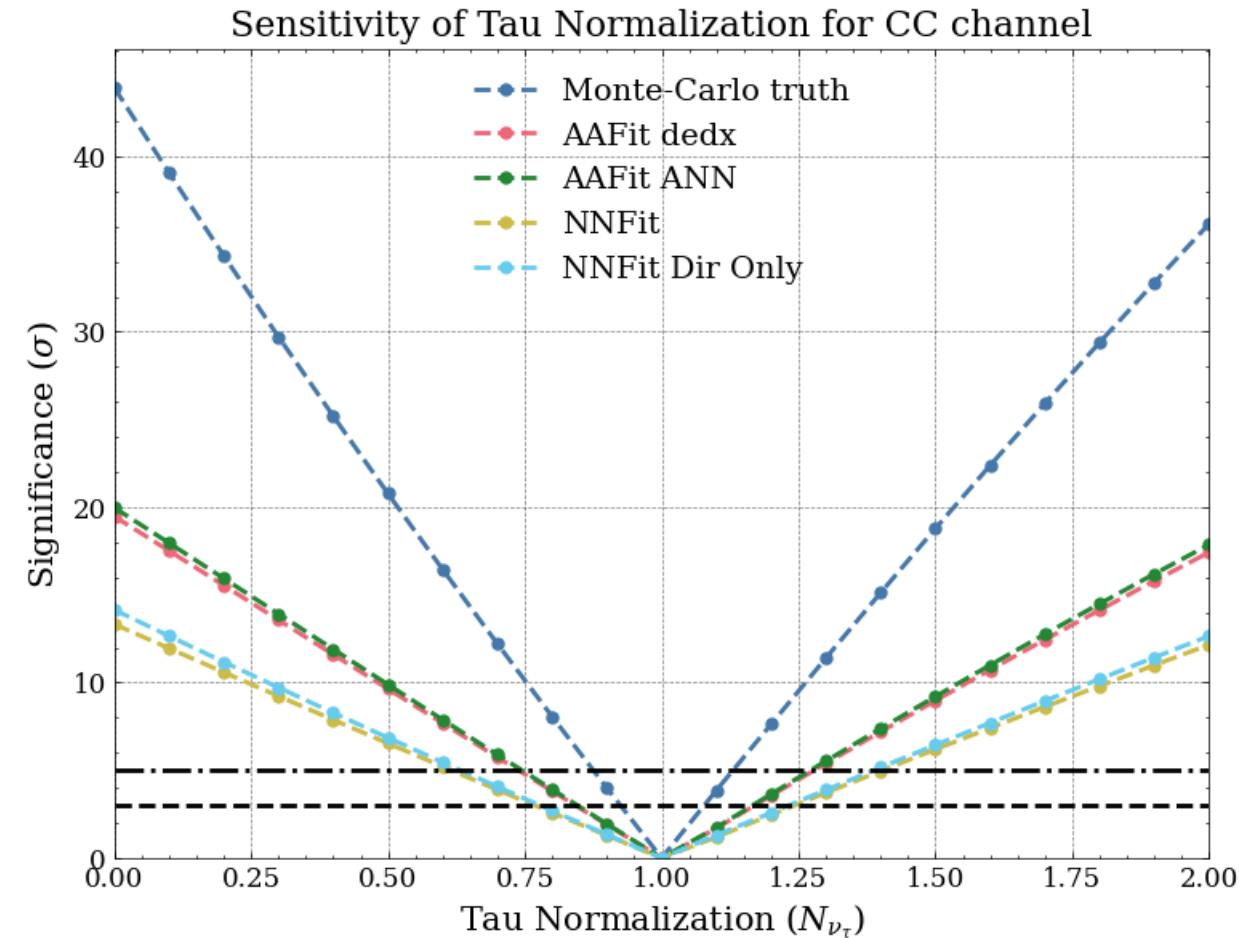
Reconstruction Sensitivities



Scenario chosen:

- "True energy" event sample
- "Charged-current" measurement
- Mass ordering: normal
- **Systematics: fixed**

Reconstruction	Energy	Direction
AAFit	dedX	aafit
AAFit	ANN	aafit
NNFit	NNFit	NNFit
NNFit Dir Only	true	NNFit
MC	true	true



Tau Appearance - Sensitivity

Reconstruction Sensitivities



Scenario chosen:

- "True energy" event sample
- "Charged-current" measurement
- Mass ordering: normal
- **Systematics: fitted**

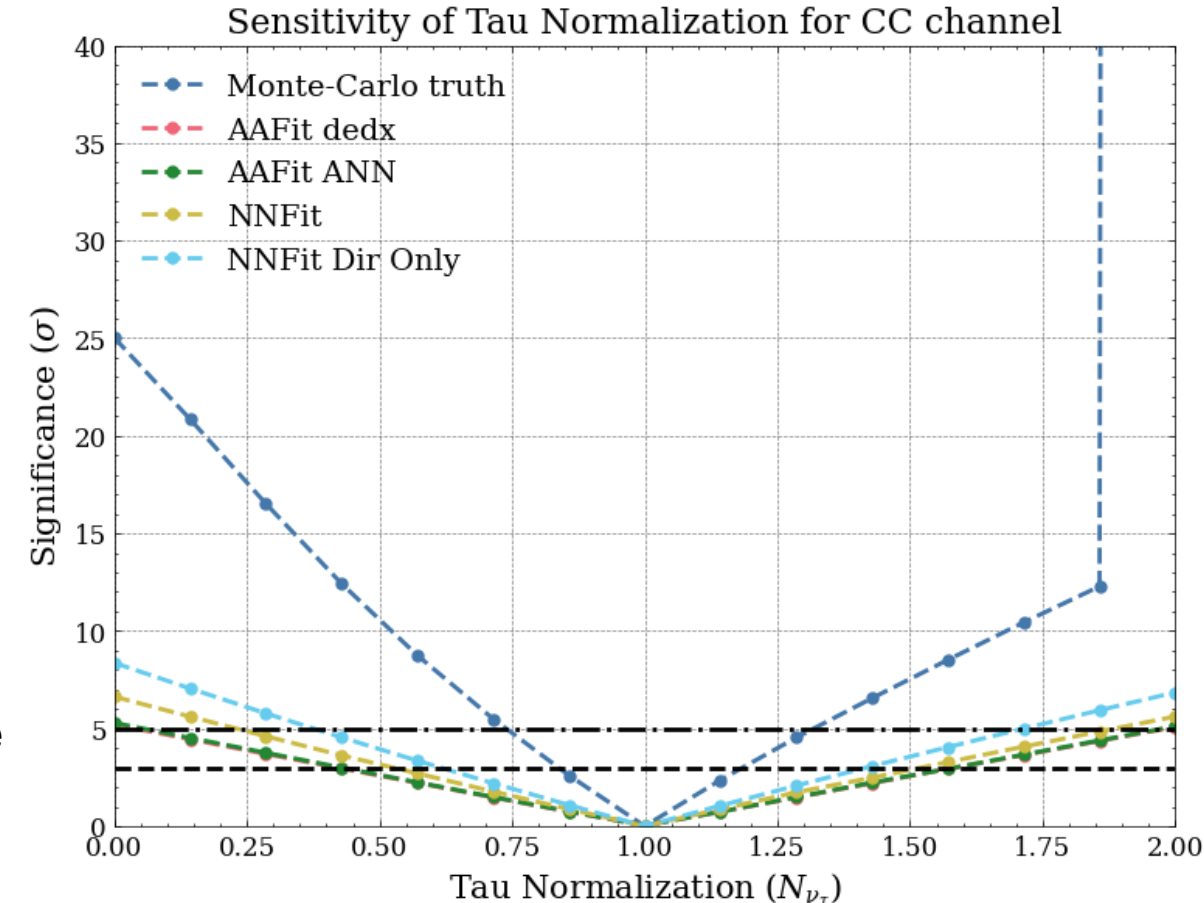
Remarks:

Same treatment as the KM3NeT study

- NNFit exhibits the best performance
- AAFit: both energy estimators have similar performance

Hypothesis - 1st clue

There is a correlation between the variance of the reconstruction performance and the way the systematics are treated!



Disclaimer: Fit of the last point failed!

Tau Appearance - Sensitivity

Reconstruction Sensitivities

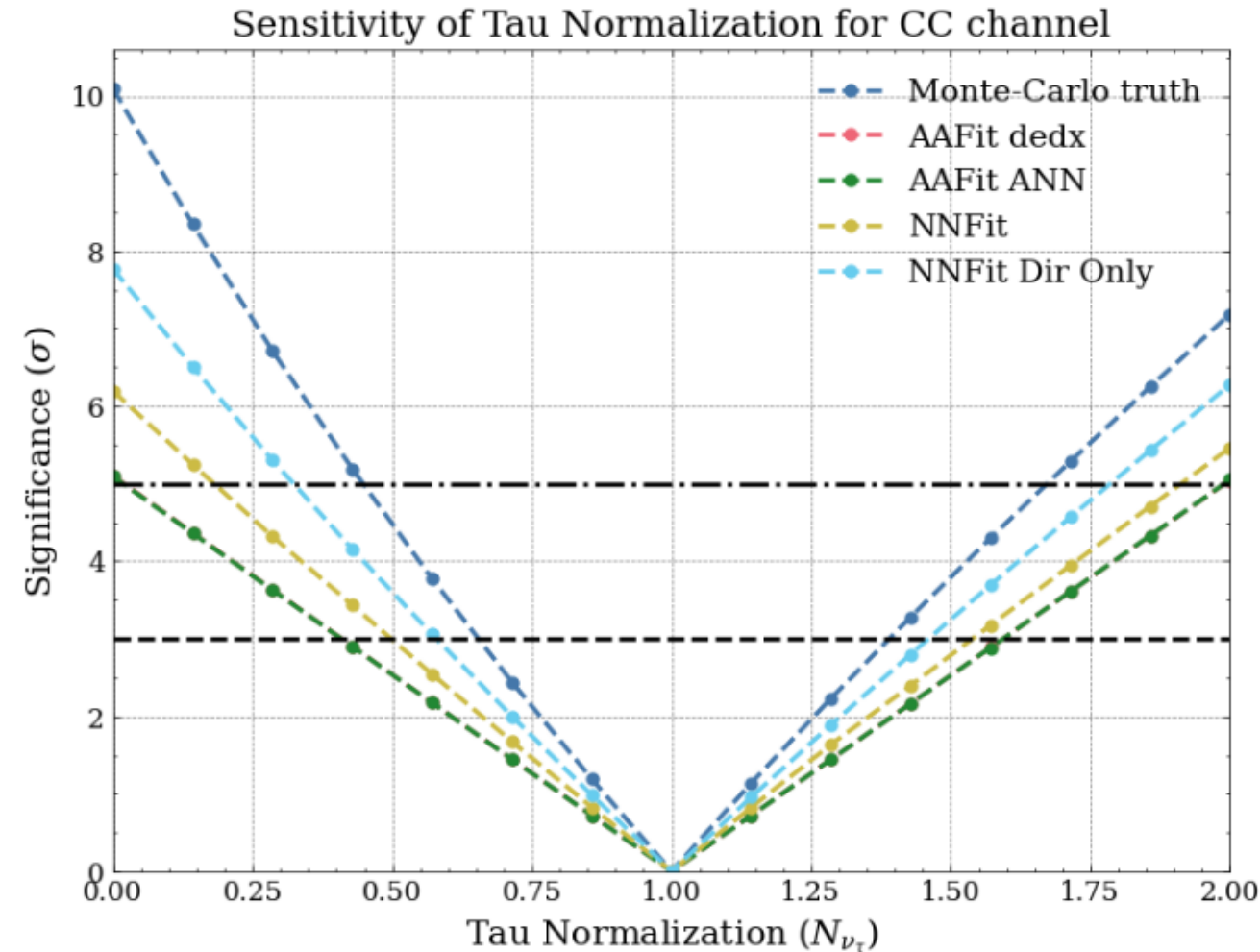


Scenario chosen:

- NNFit reconstructed events with quality cuts
- "Charged-current" measurement
- Mass ordering: normal
- Systematics: fitted

Remarks:

- NNFit reconstruction can provide up to 6σ
- Loss of 2σ due to imperfect energy reconstruction



Next step: Smearing the MC-truth events

- **Goal:** Find at which level of smearing NNFit reconstruction currently stands
- **Two Methods:**
 - "constant" smearing: smearing events proportional to their true values
 - "resolution" smearing: smearing that mimics the directional & energy resolution of NNFit

Smearing Options:

- 10% smearing
 - 30% smearing
 - 100% smearing
- ANTARES
 - ORCA 6 DUs
 - ORCA 115 DUs

Comparison with median
performance of ORCA

Tau Appearance - Sensitivity

Smearing Sensitivity - Systematic fixed scenario



ERLANGEN CENTRE
FOR ASTROPARTICLE
PHYSICS



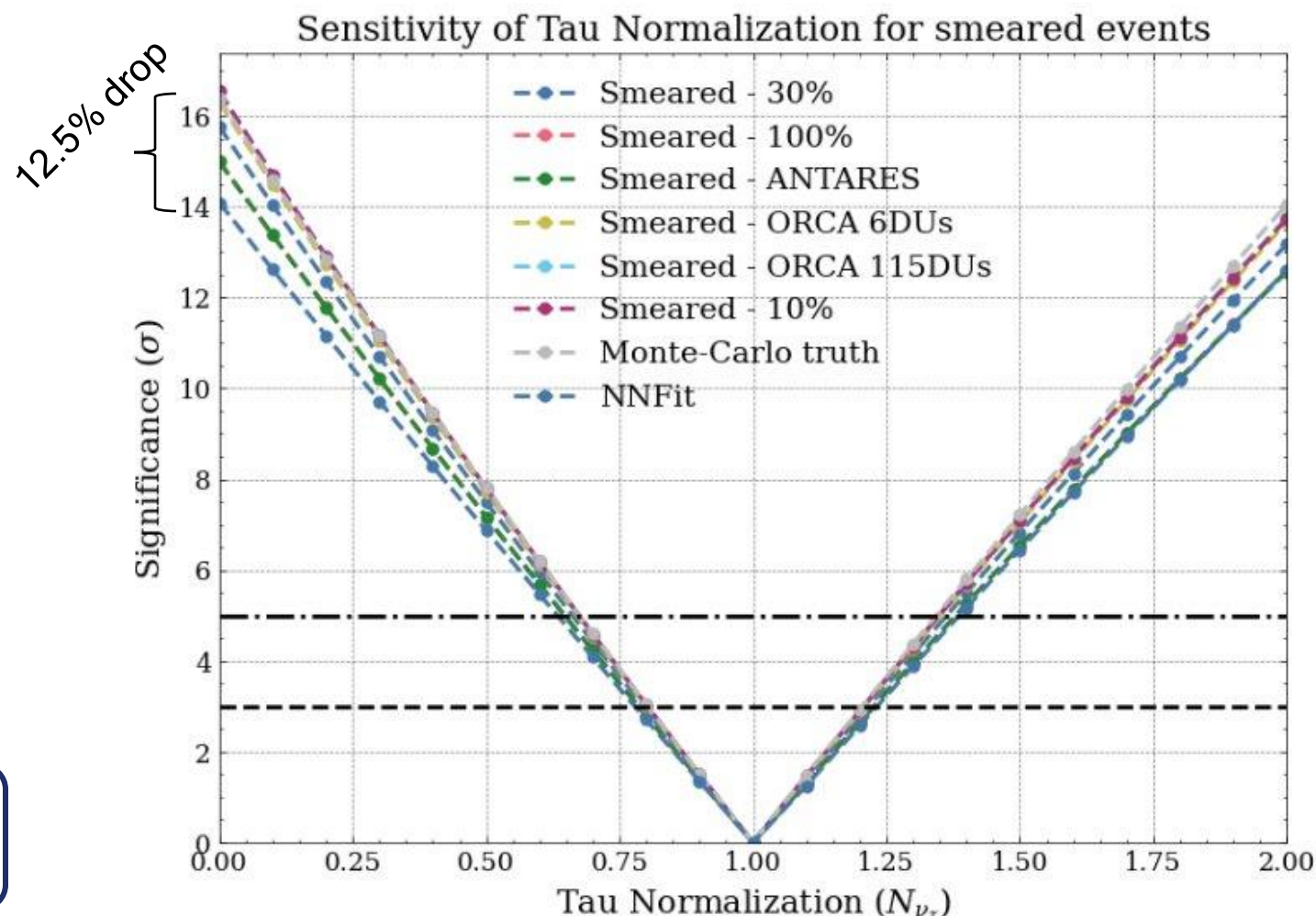
Remarks:

- Best performing scenarios very close with the unsmeared events
- No improvement detected between ORCA 6 & 115 DUs
- Current "resolution" level almost the same as 100% constant smearing

Why NNFit and Smeared-ANTARES don't have the same sensitivity?



NNFit has a significant variance in the energy reconstruction!



Tau Appearance - Sensitivity

Smearing Sensitivity - Systematics fitted scenario



ERLANGEN CENTRE
FOR ASTROPARTICLE
PHYSICS

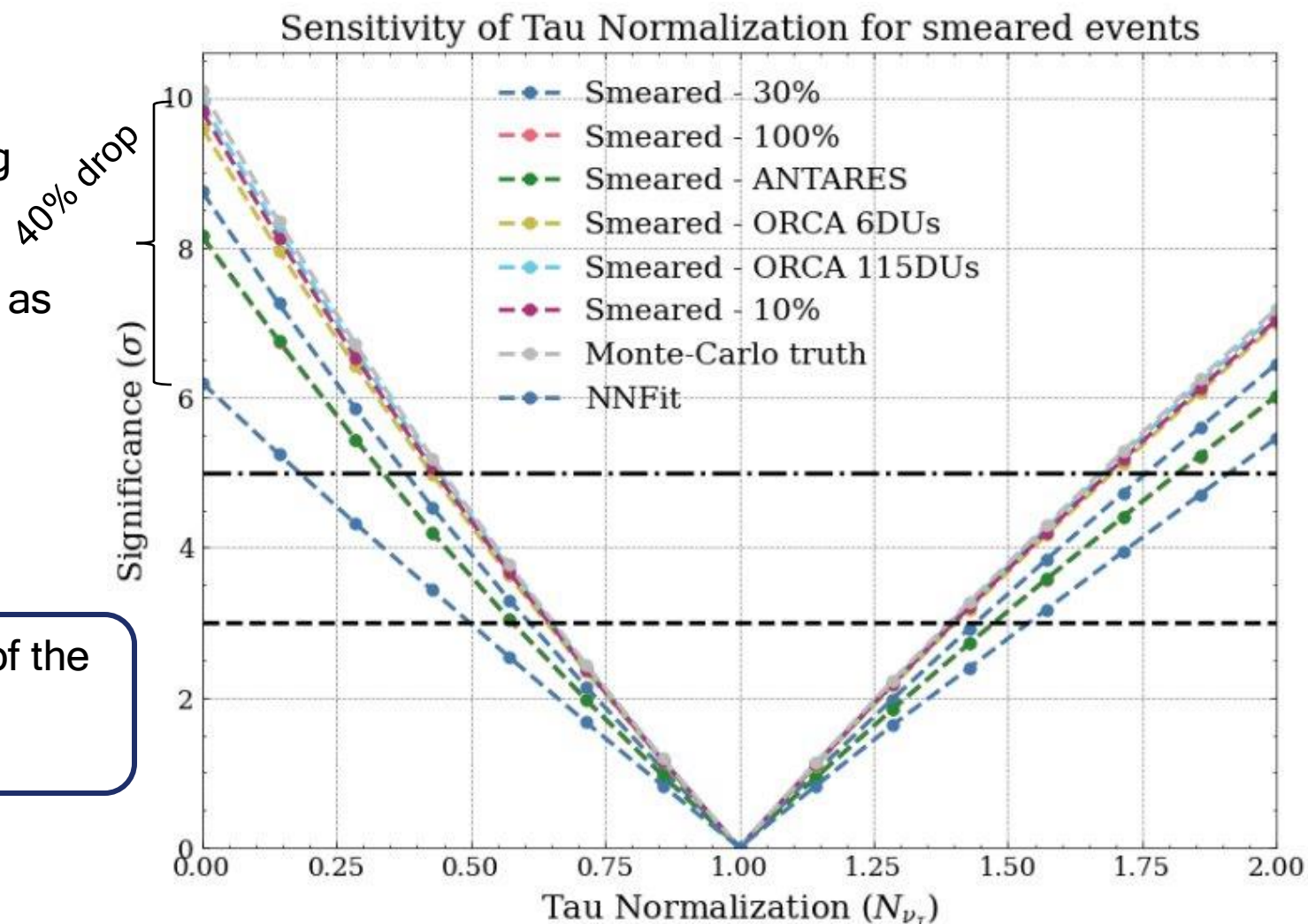


Remarks:

- Slight differences between best performing scenarios detected!
- Current "resolution" level almost the same as 100% "constant" smearing

Hypothesis - 2nd clue

There is a correlation between the variance of the reconstruction performance and the way the systematics are treated!



Tau Appearance - Sensitivity

Reconstruction algorithm improvement



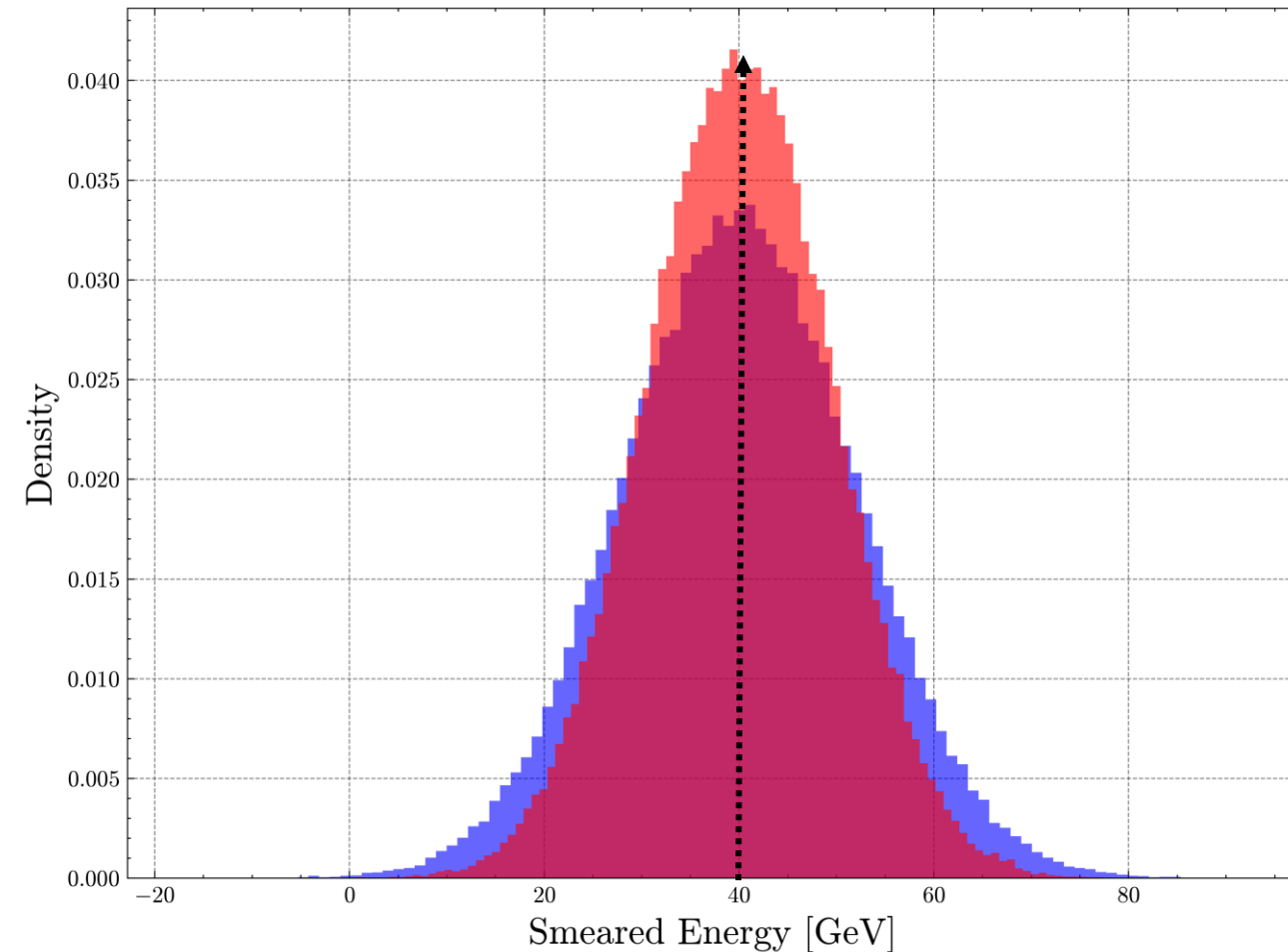
Investigation of a potential improvement to an algorithm with performance comparable to NNFit

Potential improvements investigated:

- 40% on the energy reconstruction
- 40% on the direction reconstruction
- 20% on both tasks

Strategy is similar to the "constant"- based smearing method!

- Original density distribution
- Improved density distribution



Tau Appearance - Sensitivity

Reconstruction algorithm improvement

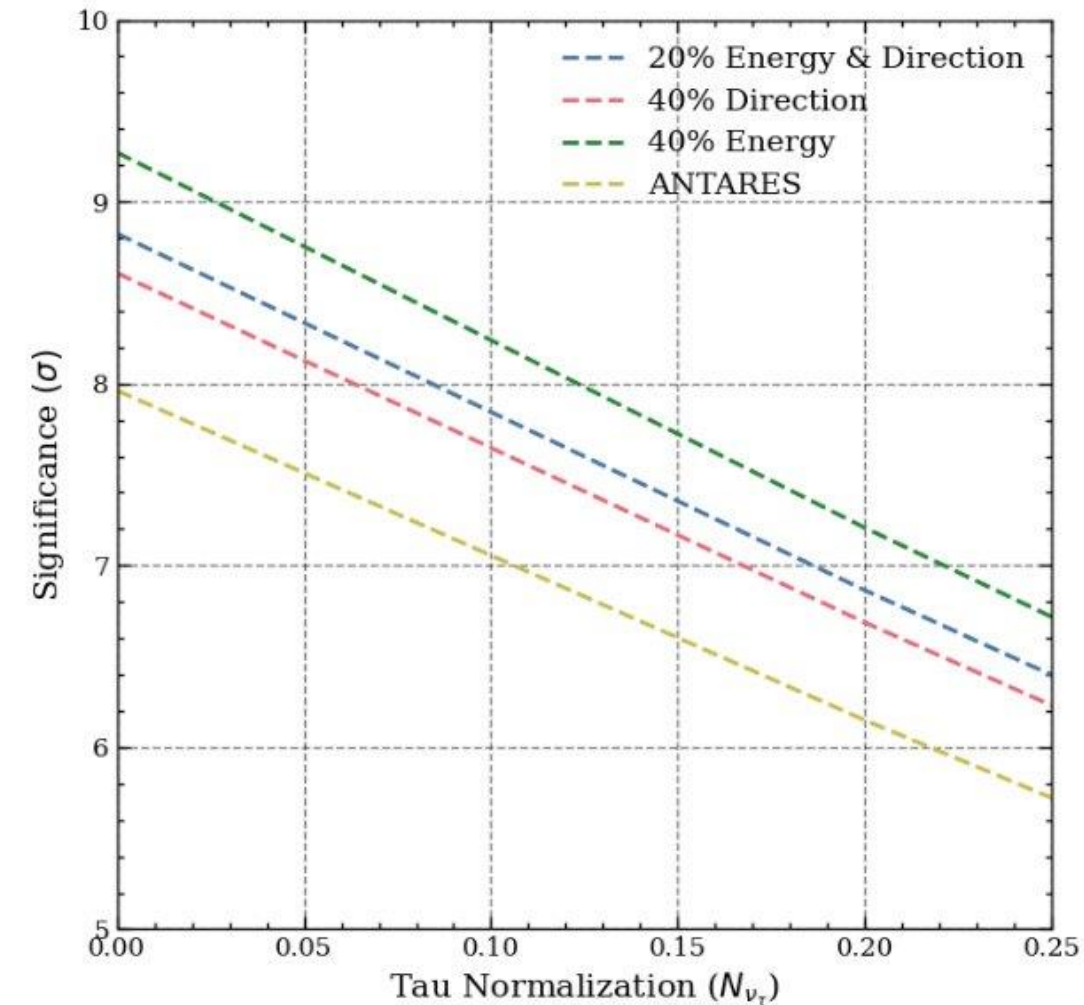


Scenario:

- Systematic treatment: fitted scenario
- Cut selection: NNFit reconstructed with quality cuts
➡ 15% of the sample

Remarks:

- Energy reconstruction task offers the biggest improvement
- Less improvement in both tasks still preforms better than direction alone



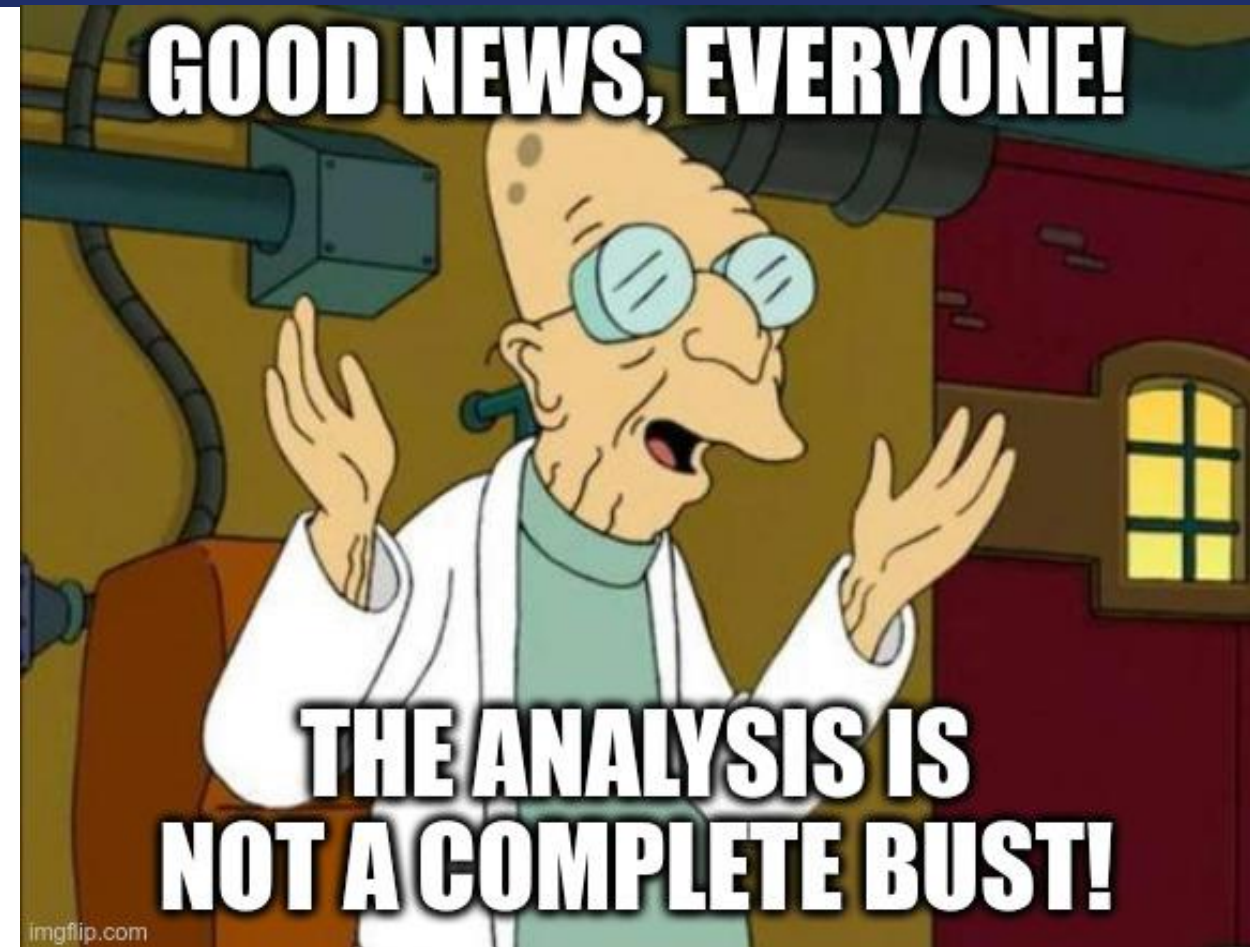
- **Processing pipeline developed for MC ANTARES files**
 - 300k AntDST files processed
 - Full sample from the entire detector sample offered
 - First pipeline to include NNFit for the purpose of *Swim*

- **Reconstruction performance**
 - NNFit not optimized for these energies!
 - Proposed focus: the energy reconstruction task
 - Potential improvement of 0.6 up to 1.5σ

- **Tau appearance study with ANTARES**
 - First sensitivity analysis with ANTARES
 - Most suitable reconstruction: NNFit
 - With "perfect" classification and muon rejection, the non-tau scenario can be excluded with a 6.6σ

- **Open Question:**
 - Is there an underlying influence between the reconstruction performance and the treatment of systematics?

- **Potential improvement**
 - Enlarge the event sample by combining BBFit & NNFit reconstructions
 - Optimize cut selection criteria & binning
- **Make the analysis more realistic**
 - Use of newly developed classifier based on NNFit
 - Study differences in parameters specific to the detector, compared to the KM3NeT/ORCA case



Thank you for your time!



Thank you very much for your attention!



Back-Up Slides

NuFIT 5.3 (2024)

without SK atmospheric data		Normal Ordering (best fit)		Inverted Ordering ($\Delta\chi^2 = 2.3$)	
		bfp $\pm 1\sigma$	3σ range	bfp $\pm 1\sigma$	3σ range
	$\sin^2 \theta_{12}$	$0.307^{+0.012}_{-0.011}$	$0.275 \rightarrow 0.344$	$0.307^{+0.012}_{-0.011}$	$0.275 \rightarrow 0.344$
	$\theta_{12}/^\circ$	$33.66^{+0.73}_{-0.70}$	$31.60 \rightarrow 35.94$	$33.67^{+0.73}_{-0.71}$	$31.61 \rightarrow 35.94$
	$\sin^2 \theta_{23}$	$0.572^{+0.018}_{-0.023}$	$0.407 \rightarrow 0.620$	$0.578^{+0.016}_{-0.021}$	$0.412 \rightarrow 0.623$
	$\theta_{23}/^\circ$	$49.1^{+1.0}_{-1.3}$	$39.6 \rightarrow 51.9$	$49.5^{+0.9}_{-1.2}$	$39.9 \rightarrow 52.1$
	$\sin^2 \theta_{13}$	$0.02203^{+0.00056}_{-0.00058}$	$0.02029 \rightarrow 0.02391$	$0.02219^{+0.00059}_{-0.00057}$	$0.02047 \rightarrow 0.02396$
	$\theta_{13}/^\circ$	$8.54^{+0.11}_{-0.11}$	$8.19 \rightarrow 8.89$	$8.57^{+0.11}_{-0.11}$	$8.23 \rightarrow 8.90$
	$\delta_{CP}/^\circ$	197^{+41}_{-25}	$108 \rightarrow 404$	286^{+27}_{-32}	$192 \rightarrow 360$
	$\frac{\Delta m_{21}^2}{10^{-5} \text{ eV}^2}$	$7.41^{+0.21}_{-0.20}$	$6.81 \rightarrow 8.03$	$7.41^{+0.21}_{-0.20}$	$6.81 \rightarrow 8.03$
	$\frac{\Delta m_{3\ell}^2}{10^{-3} \text{ eV}^2}$	$+2.511^{+0.027}_{-0.027}$	$+2.428 \rightarrow +2.597$	$-2.498^{+0.032}_{-0.024}$	$-2.581 \rightarrow -2.409$

with SK atmospheric data		Normal Ordering (best fit)		Inverted Ordering ($\Delta\chi^2 = 9.1$)	
		bfp $\pm 1\sigma$	3σ range	bfp $\pm 1\sigma$	3σ range
	$\sin^2 \theta_{12}$	$0.307^{+0.012}_{-0.011}$	$0.275 \rightarrow 0.344$	$0.307^{+0.012}_{-0.011}$	$0.275 \rightarrow 0.344$
	$\theta_{12}/^\circ$	$33.67^{+0.73}_{-0.71}$	$31.61 \rightarrow 35.94$	$33.67^{+0.73}_{-0.71}$	$31.61 \rightarrow 35.94$
	$\sin^2 \theta_{23}$	$0.454^{+0.019}_{-0.016}$	$0.411 \rightarrow 0.606$	$0.568^{+0.016}_{-0.021}$	$0.412 \rightarrow 0.611$
	$\theta_{23}/^\circ$	$42.3^{+1.1}_{-0.9}$	$39.9 \rightarrow 51.1$	$48.9^{+0.9}_{-1.2}$	$39.9 \rightarrow 51.4$
	$\sin^2 \theta_{13}$	$0.02224^{+0.00056}_{-0.00057}$	$0.02047 \rightarrow 0.02397$	$0.02222^{+0.00069}_{-0.00057}$	$0.02049 \rightarrow 0.02420$
	$\theta_{13}/^\circ$	$8.58^{+0.11}_{-0.11}$	$8.23 \rightarrow 8.91$	$8.57^{+0.13}_{-0.11}$	$8.23 \rightarrow 8.95$
	$\delta_{CP}/^\circ$	232^{+39}_{-25}	$139 \rightarrow 350$	273^{+24}_{-26}	$195 \rightarrow 342$
	$\frac{\Delta m_{21}^2}{10^{-5} \text{ eV}^2}$	$7.41^{+0.21}_{-0.20}$	$6.81 \rightarrow 8.03$	$7.41^{+0.21}_{-0.20}$	$6.81 \rightarrow 8.03$
	$\frac{\Delta m_{3\ell}^2}{10^{-3} \text{ eV}^2}$	$+2.505^{+0.024}_{-0.026}$	$+2.426 \rightarrow +2.586$	$-2.487^{+0.027}_{-0.024}$	$-2.566 \rightarrow -2.407$

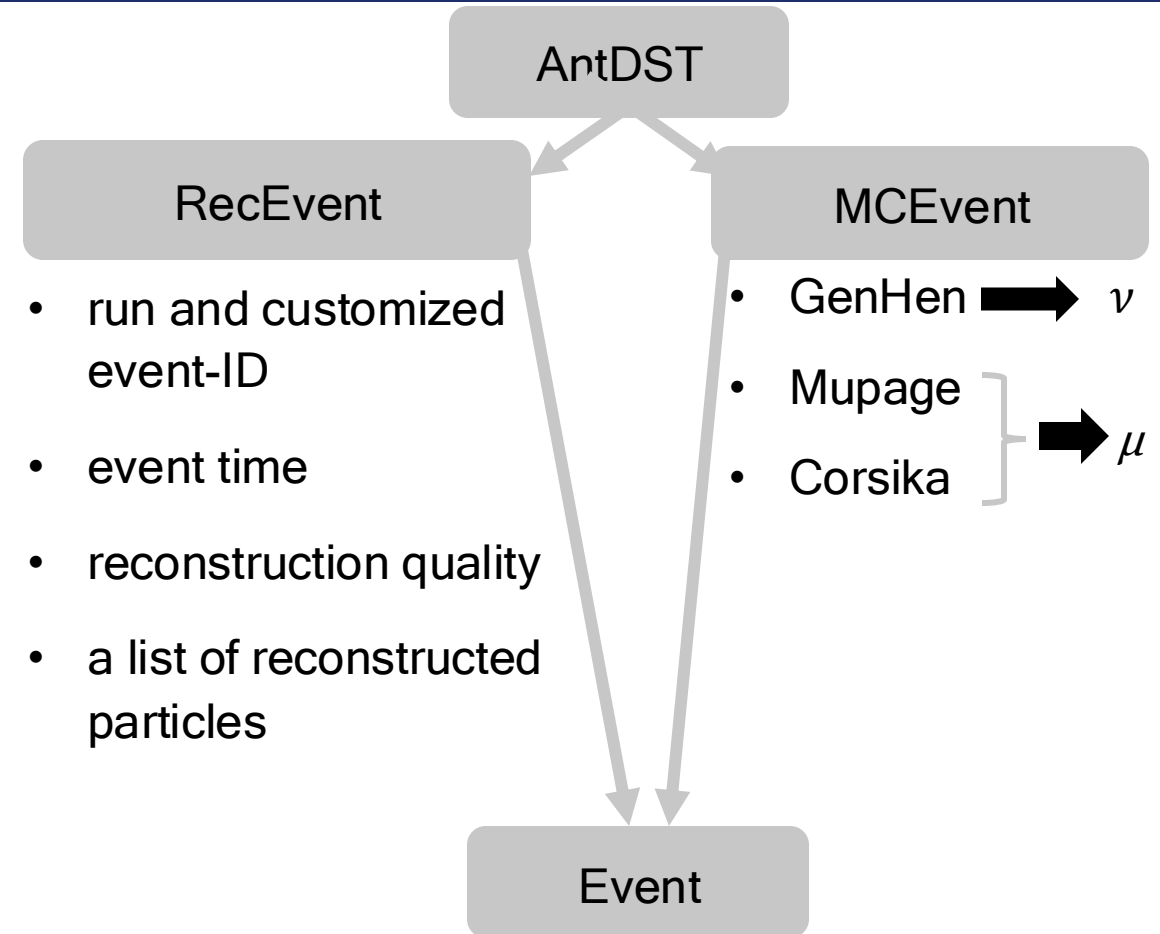
- Final stage files of ANTARES Monte-Carlo chain
- Contains **only** high-level variables

- run information
- reconstructed information
- "true" information

Aim: Suitable for a physics analysis!

- Summary

➡ For each run there are $8\nu_e$, $8\nu_\mu$, $12\nu_\tau$ and one μ files



- Naming convention:

Particle type:

- neutrinos
 - (a)nue
 - (a)numu
 - (a)nutau
- muons

Run-ID

Example: MC_068813_numu_a_CC_reco.root

Energy range:

- a for 5 GeV to 20 TeV
- b for 20 TeV to 100 PeV

Interaction type:

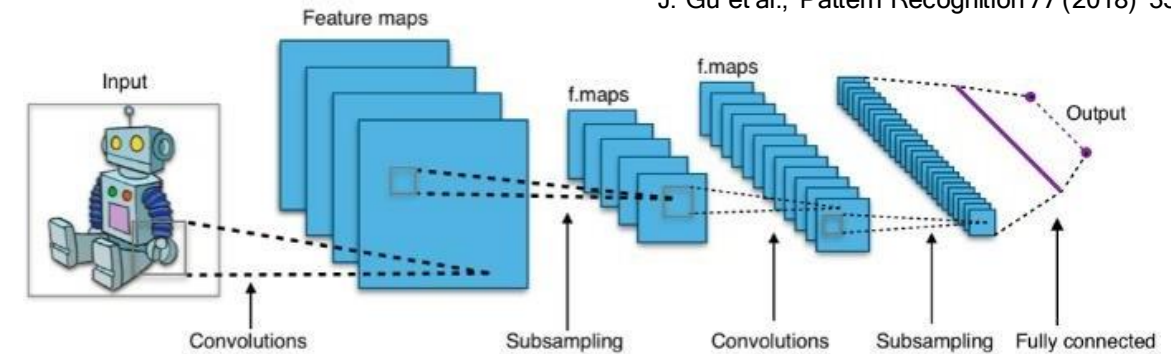
- CC • CCmu
- NC • CCshow

Attention: Second row only for tau neutrinos!

- Summary

➡ For each run there are $8\nu_e$, $8\nu_\mu$, $12\nu_\tau$ and one μ files

- Deep learning reconstruction method for ANTARES
 - Deep Convolutional Networks (DCNs) + Mixture Density Networks(MDNs)
- Developed for low-energy events ($E_{\text{true}} < 150 \text{ GeV}$)
- Single-line event detection
- 2 models provided
 - NNFitTrack
 - NNFitShower
- Reco Variables:
 - Zenith, azimuth, energy, interaction vertex



Garcia-Mendez, J., Ardid, S., & Ardid, M. (2023). Dark matter search towards the Sun using Machine Learning reconstructions of single-line events in ANTARES. PoS, ICRC2023, 1443.

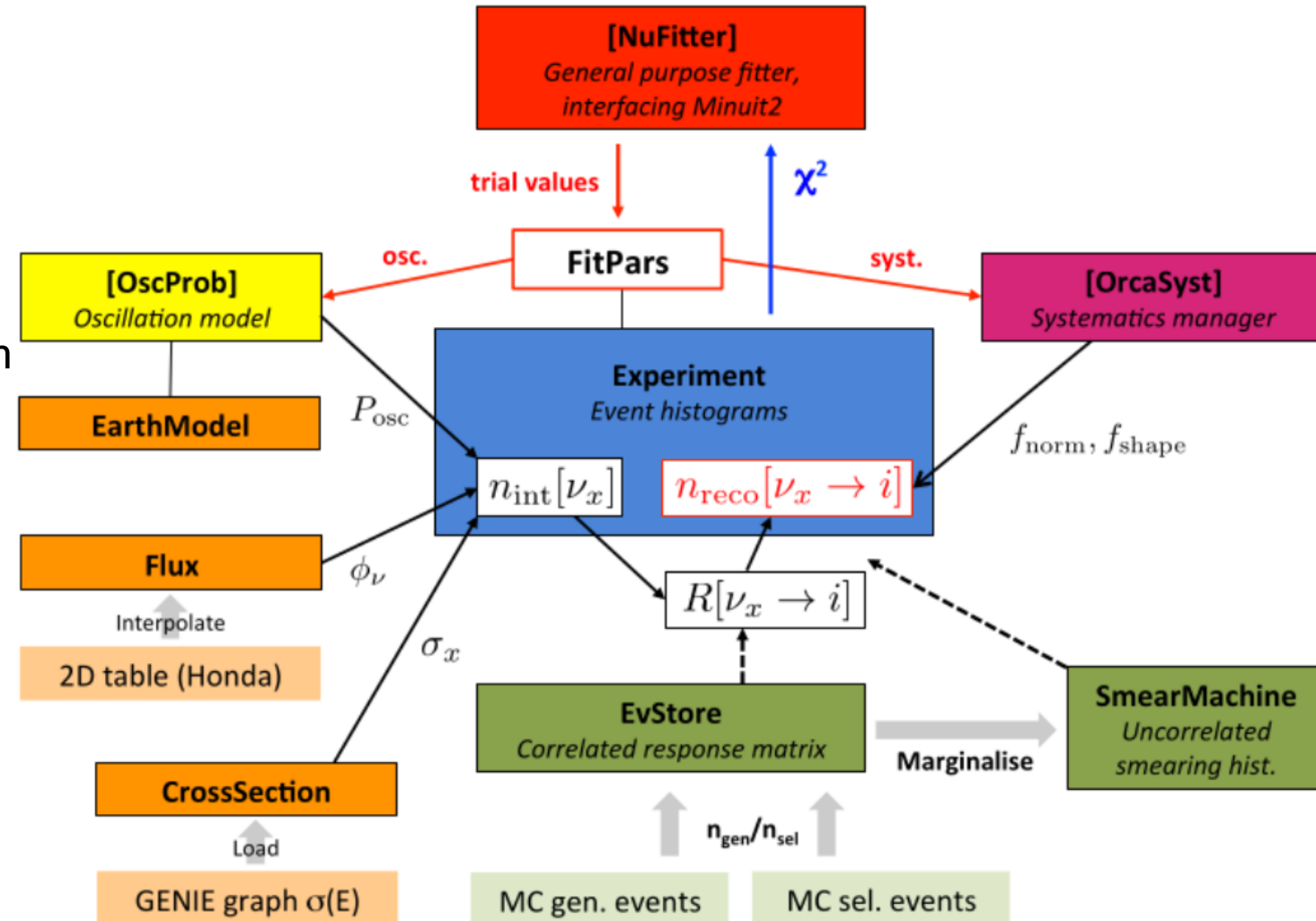
Tau Appearance - Sensitivity

Swim software



Swim:

- Internal software for the oscillation WG of KM3NeT
- Framework based on ROOT & C++
- Provides methods for sensitivity calculation in various analyses (SM & BSM)
- Main packages:
 - OrcaSim: Simulate experiments
 - OscProb: Oscillation probabilities
 - OrcaDet: Simulate detector response
 - NuFitter: Minimization class (Minuit)
 - OrcaSyst: Systematics handler



Tau Appearance - Sensitivity

Analysis systematics



Systematic fixed scenario

Parameter	Initial value	Prior
θ_{12}	33.44 (NO) 33.45 (IO)	fixed
θ_{13}	8.57 (NO) 8.60 (IO)	fixed
θ_{23}	49.2 (NO) 49.3 (IO)	fixed
Δm_{31} [eV ²]	2.517e-03 (NO) -2.4238e-03 (IO)	fixed
Δm_{21} [eV ²]	7.42e-05	fixed
δ_{CP}	197.0 (NO) 282.0 (IO)	fixed
Energy Slope	0	10%
Zenith Slope	0	2%
ν_{μ} skew	0	5%
ν_e skew	0	7%
flavor skew	0	2%
NC scale	1	20%
Energy scale	1	9%
FluxNorm	1	free
TrackNorm	1	fixed
ShowerNorm	1	fixed
MuonNorm	1	fixed

Systematics fitted scenario

Parameter	Initial value	Prior
θ_{12}	33.44 (NO) 33.45 (IO)	fixed
θ_{13}	8.57 (NO) 8.60 (IO)	fixed
θ_{23}	49.2 (NO) 49.3 (IO)	free
Δm_{31} [eV ²]	2.517e-03 (NO) -2.4238e-03 (IO)	free
Δm_{21} [eV ²]	7.42e-05	fixed
δ_{CP}	197.0 (NO) 282.0 (IO)	fixed
Energy Slope	0	10%
Zenith Slope	0	2%
ν_{μ} skew	0	5%
ν_e skew	0	7%
flavor skew	0	2%
NC scale	1	20%
Energy scale	1	9%
FluxNorm	1	free
TrackNorm	1	free
ShowerNorm	1	free
MuonNorm	1	free

Tau Appearance - Sensitivity

Smearing Method - Example



ERLANGEN CENTRE
FOR ASTROPARTICLE
PHYSICS

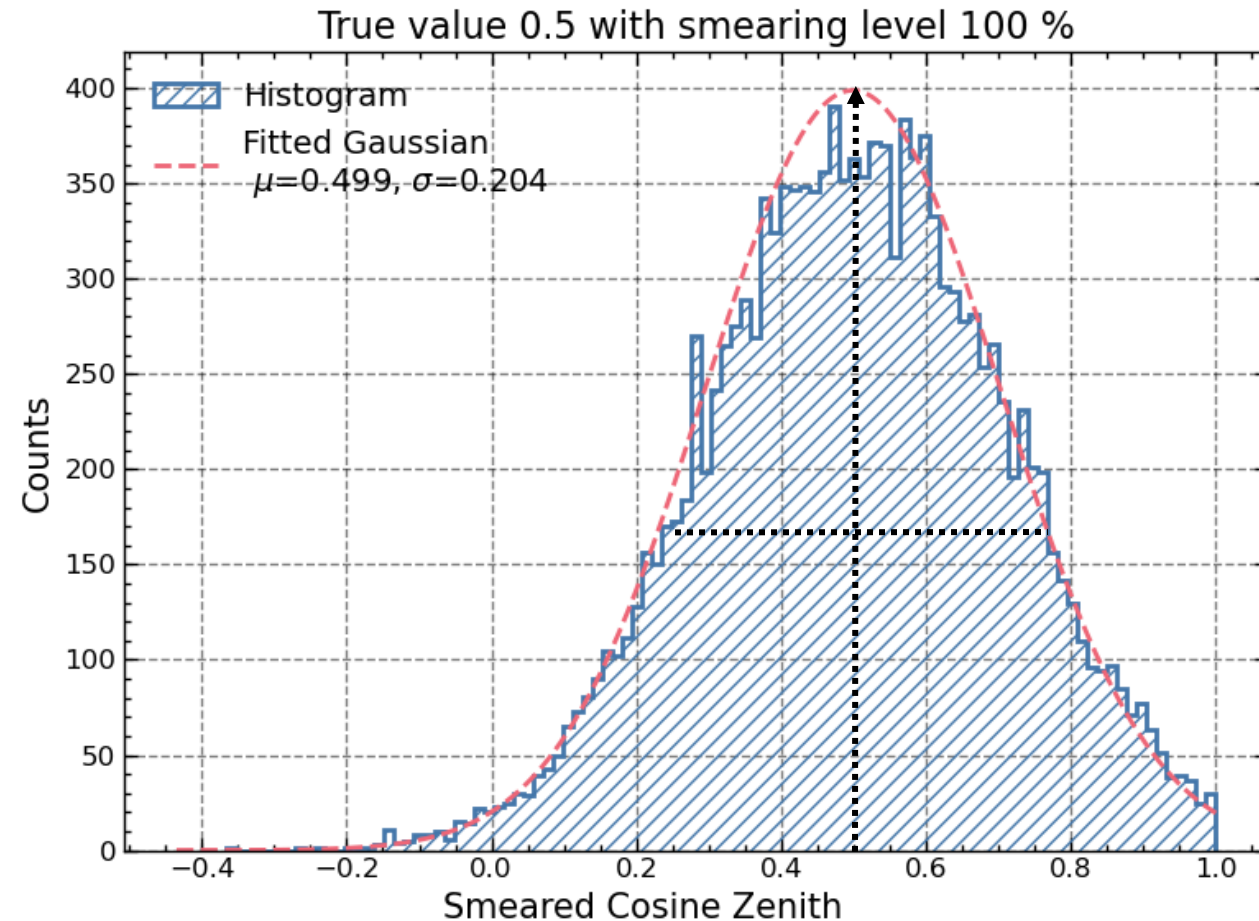


Steps:

1. Create a Gaussian with mean the true value and error proportional to the true value
2. Randomly sample a new value from the generated distribution
3. If new value, physically impossible repeat step 2

For each event smear both direction and energy branch!

Attention: in the "performance" smearing, the error is based on a function parameterizing the behavior of NNFit



Tau Appearance - Sensitivity

Smearing Method - "performance" smearing



1D parameterization of the reconstruction performance for each variable

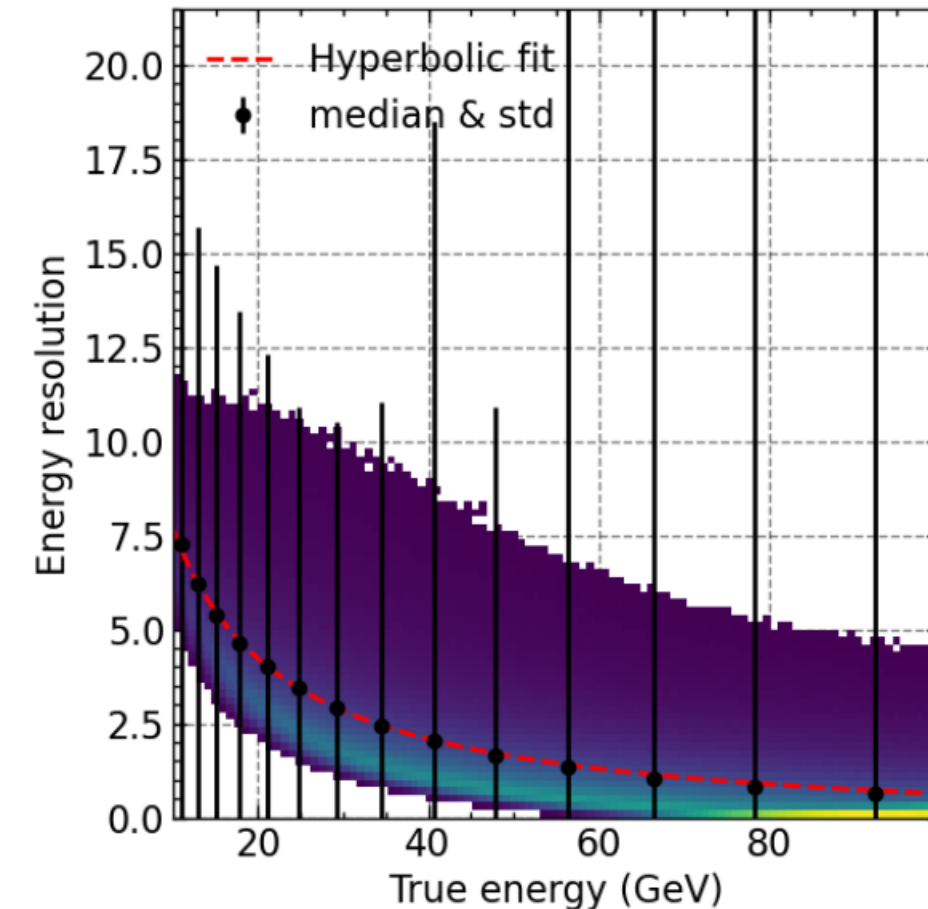
- **energy:** assumption to follow a hyperbolic behavior

$$y = \frac{\alpha}{(\beta x + \gamma)} + \delta$$

- Noticeable variance on each bin
- Potential influence of event's direction



2D approximation
not investigated



Tau Appearance - Sensitivity

Smearing Method - "performance" smearing



ERLANGEN CENTRE
FOR ASTROPARTICLE
PHYSICS



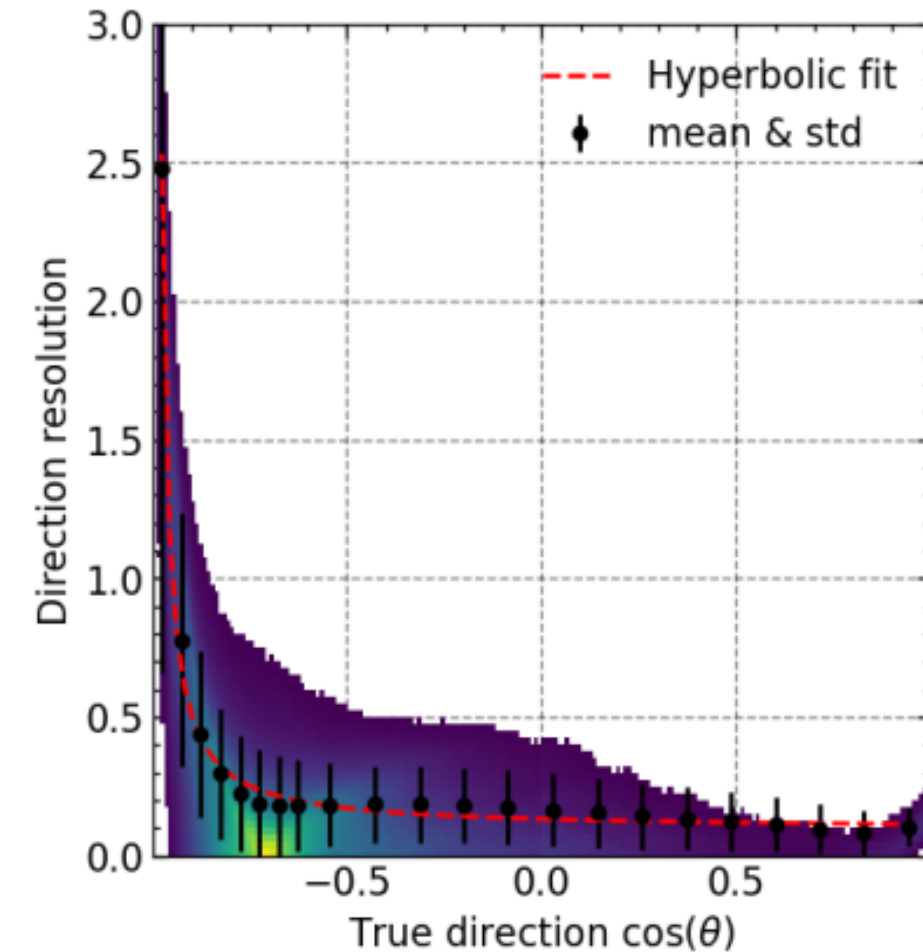
1D parameterization of the reconstruction performance for each variable

- **energy:** assumption to follow a hyperbolic behavior

$$y = \frac{\alpha}{(\beta x + \gamma)} + \delta$$

- Noticeable variance on each bin
- Potential influence of event's direction
- **direction:** hyperbolic function again can describe it
 - No influence of event's energy present
 - 90% event have a resolution less than 40%

2D approximation
not investigated

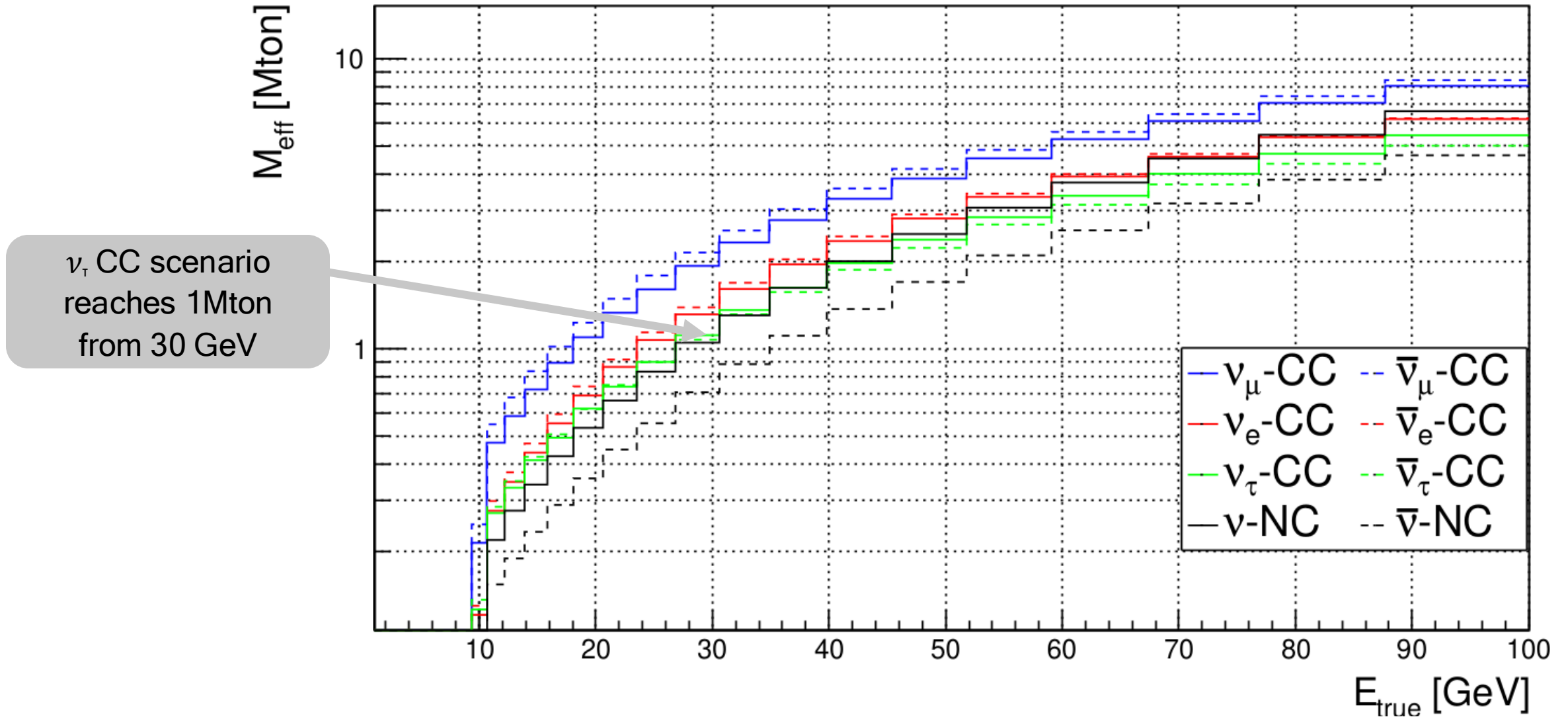


Detector effective volume

Full detector sample



ERLANGEN CENTRE
FOR ASTROPARTICLE
PHYSICS

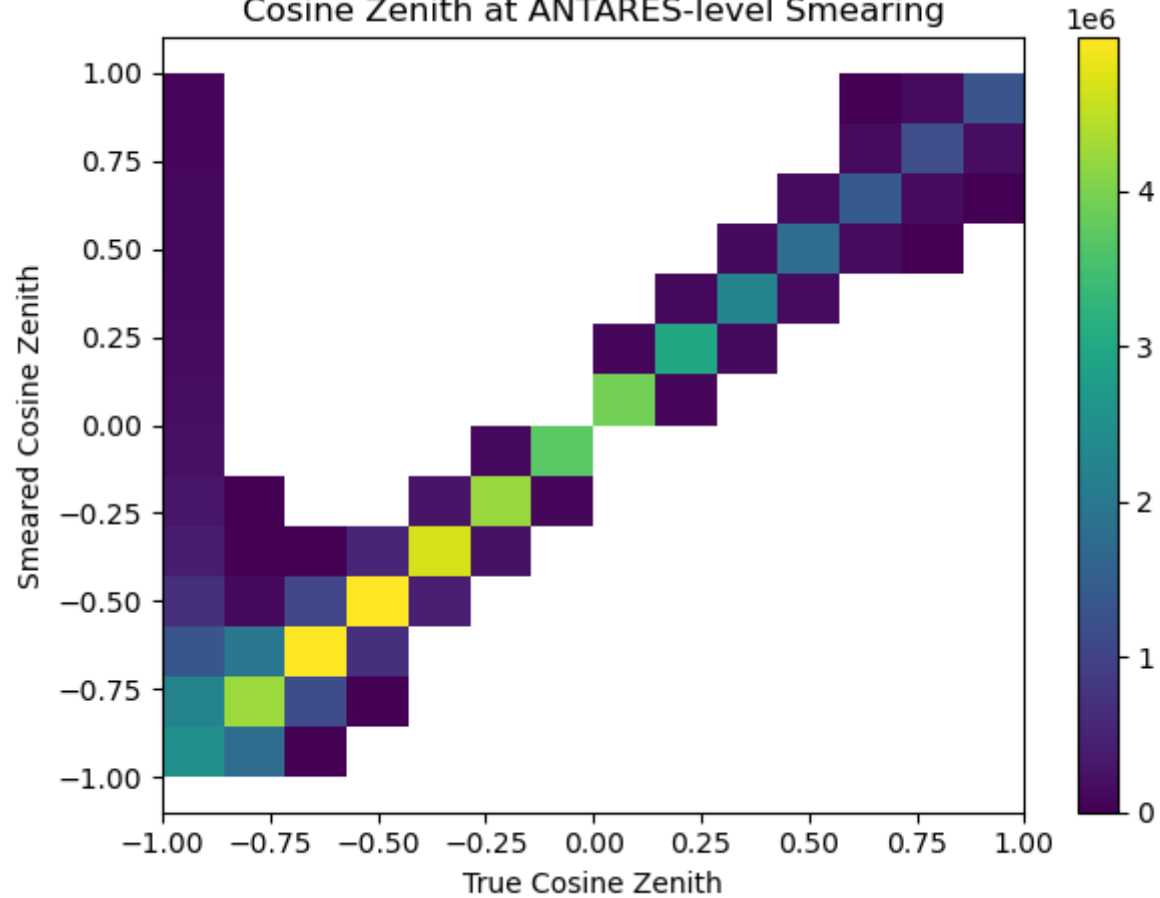


Smearing Comparison

"performance" reconstruction



Cosine Zenith at ANTARES-level Smearing



Energy at ANTARES-level Smearing

